ICMAB EXCELENCIA SEVERO OCHOA
INSTITUT DE CIÈNCIA DE MATERIALS DE BARCELONA

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SALA D'ACTES CARLES MIRAVITLLES
ICMAB-CSIC, Bellaterra, Spain

**MATENER2018**
SEVERO OCHOA SUMMER SCHOOL
ON MATERIALS FOR ENERGY

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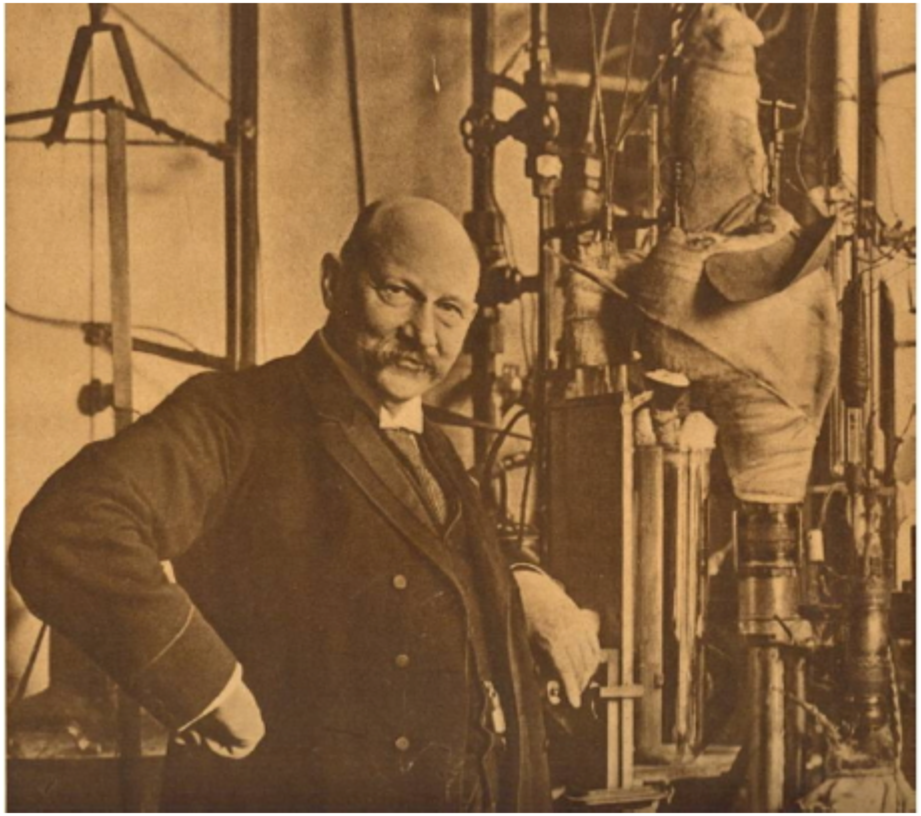
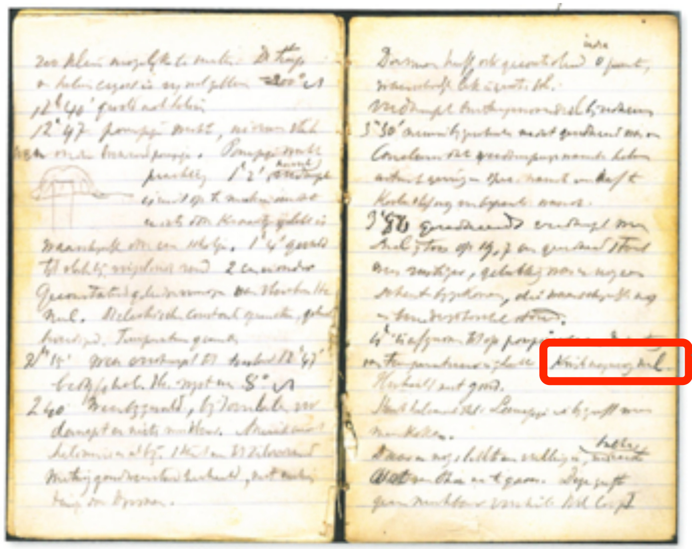
# Superconductivity II
## Large Scale Applied Superconductivity

Pascal Tixador
Grenoble-INP, Université de Grenoble Alpes, CNRS
G2ELab, Institut Néel

UNIVERSITÉ Grenoble Alpes · Grenoble INP · CNRS · NÉEL institut · G2E Lab Grenoble Génie Electrique Grenoble Electrical Engineering

# 100 years ago: Avril, 8, 1911





"Mercury nearly zero"

# Objectives and outline

The main purpose is to provide informations about applied superconductivity from phenomena to applications through materials.

- Introduction
- Historical references
- Superconductivity for large scale applications
- Superconducting material (REBCO tapes)
- Superconducting applications
- Fault Current Limiter (FCL)

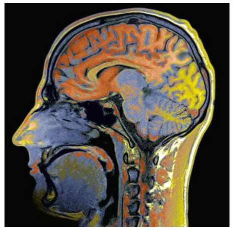
# Introduction

# Superconductivity: industrial activity

## Niche market but industrial market

- About 5 185 MEuros (worldwide market in 2012)
- **Medical imaging** (2500 MRI per year, 30 000 in service)
  - Non invasive extremely useful tool
  - ***"MRI*** (Magnetic Resonance Imaging) has transformed super-conductivity from scientific laboratory to everyday use; ***Superconductivity*** made MRI a commercial reality" M. Parizh (Phillips).
- **RMN high resolution Spectrometer RMN**
  - Indispensable and outstanding analysis tool
- Thermonuclear Fusion (ITER)
  - Solar energy for tomorrow energy
- High energy physics
  - Origins of universe (LHC)

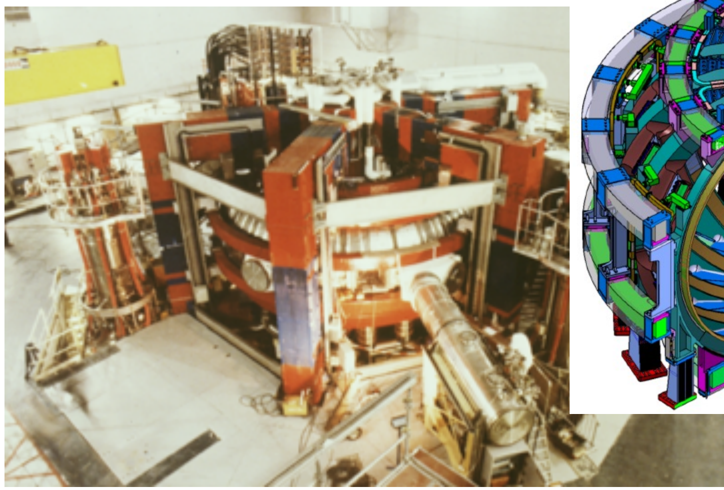
# Main superconducting applications

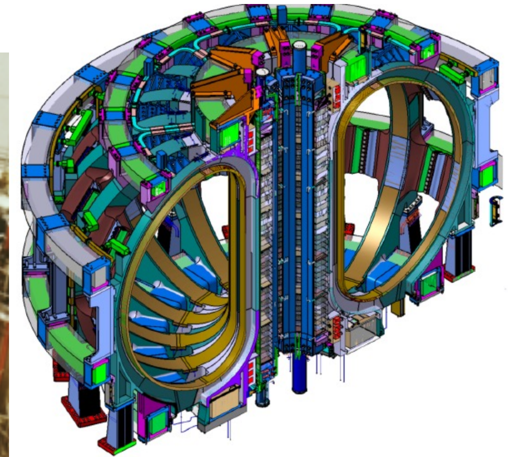


40 000
in service

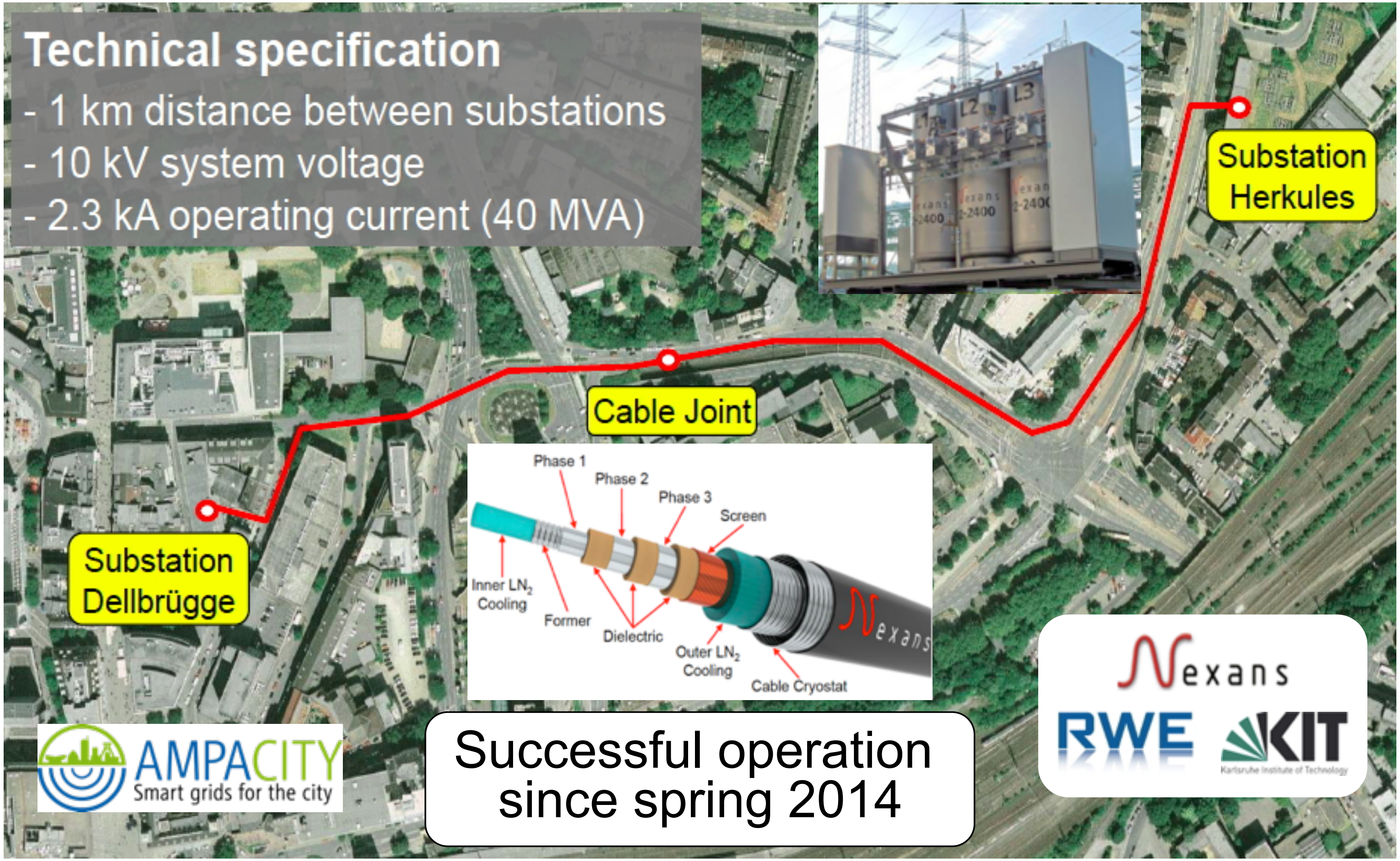
27 km
(100 km)

40 MW
(900 MW)



# Cable + FCL project: AMPACITY

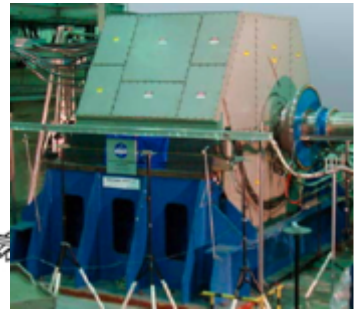



Courtesy M. Stemmle

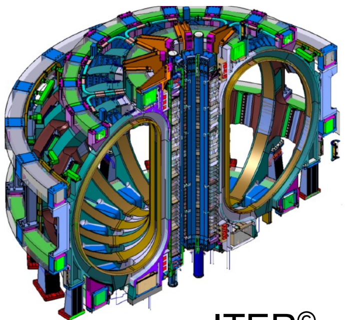
# « Superconducting » world

Siemens©

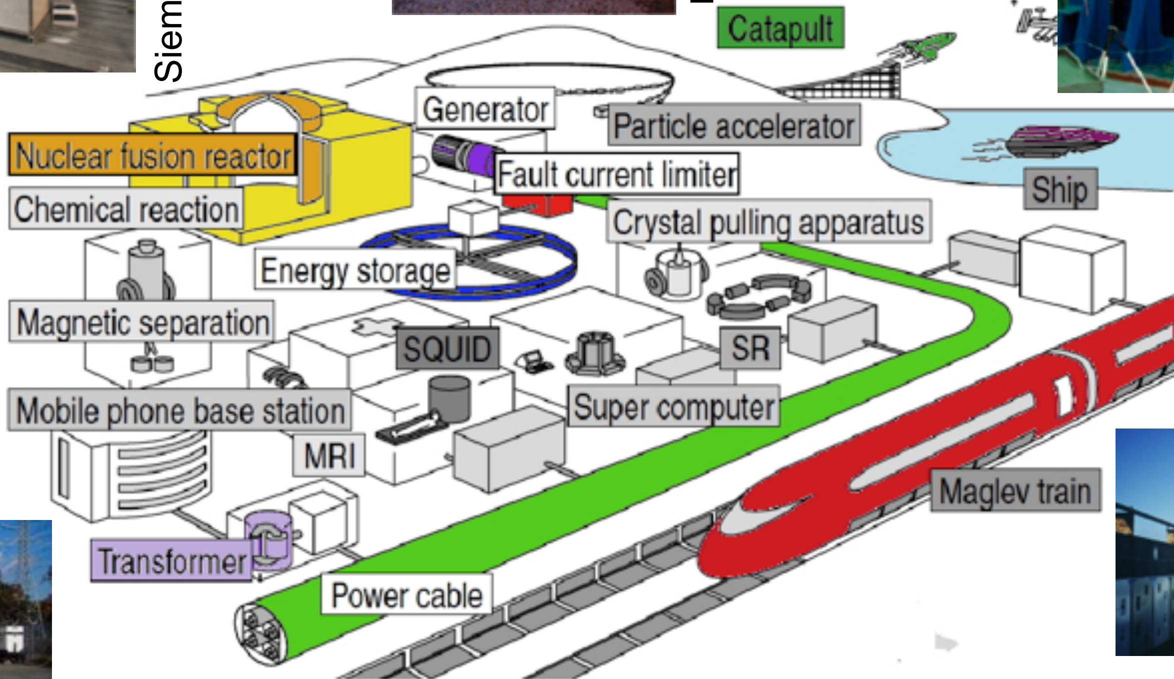
NEXANS©



AMSC©



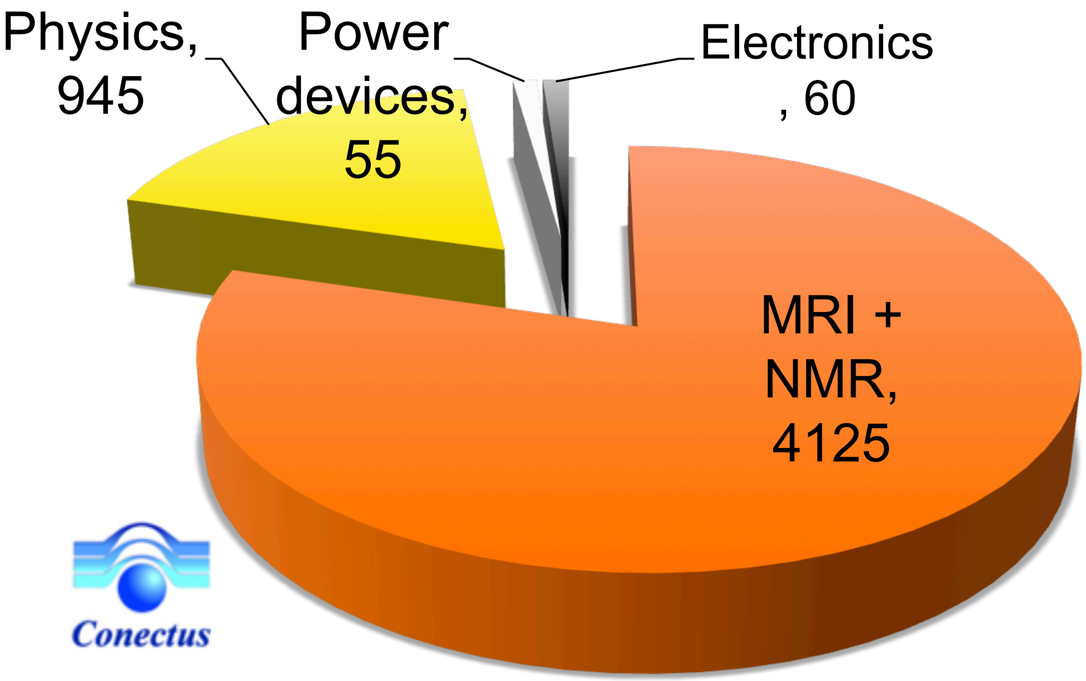
ITER©

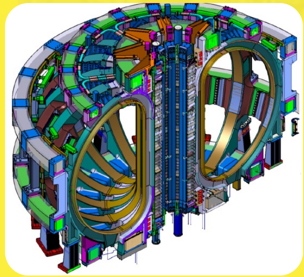



SEI technical review

NEXANS©

# Superconductivity market (2011)



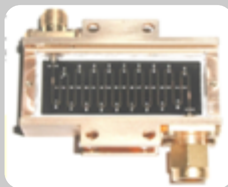



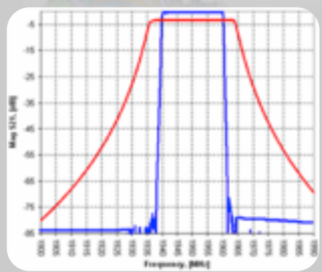
**Total : 5080 MEuros (2011)**

LTS market: 5050 M€

HTS market: 30 M€

**HTS**





# Some companies

## Big companies

- Siemens
- Oxford Instruments Superconductivity
- GE
- Philips
- NEXANS
- SuperPower Furikawa
- …

## SME and start ups

- THEVA
- Oxolutia
- …

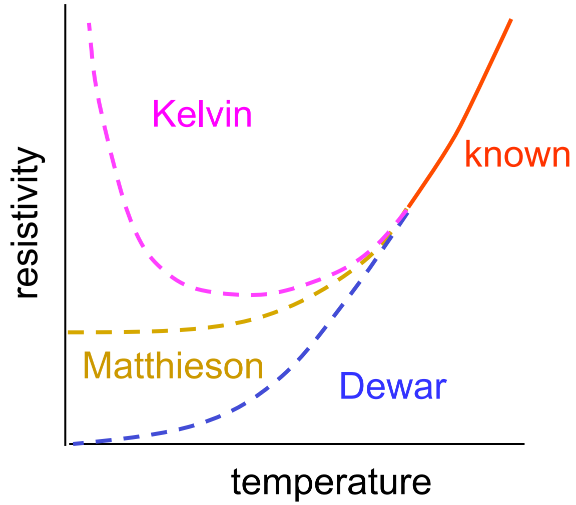
# Some historical references

# Beginning of the story

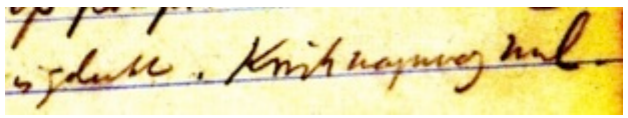
Once in Leiden …

# Superconductivity discovery

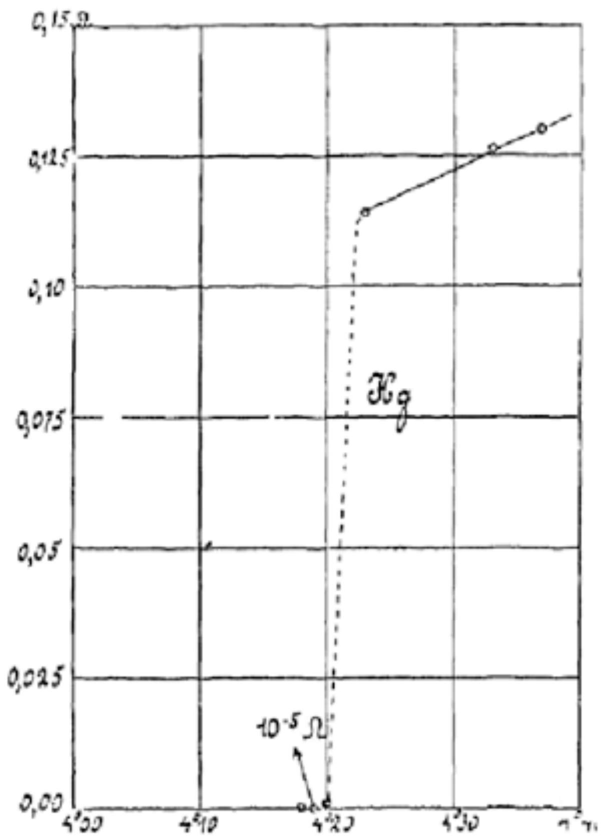
- Metal resistivity studies

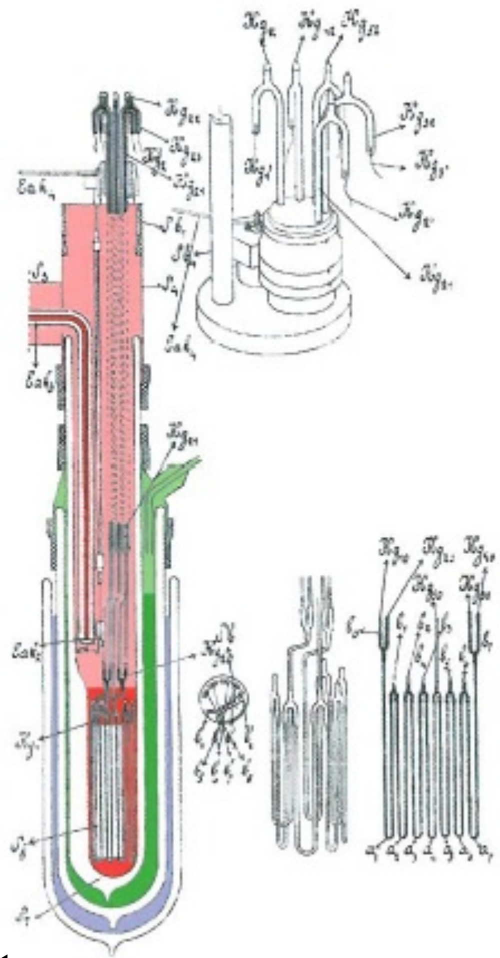



**April 8, 1911**



*Heike Kammerlingh Onnes*
*Notebook #56*



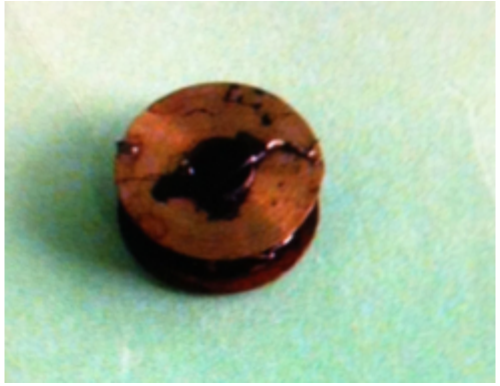



**October 26, 1911** **G. Holst**

# Big hopes

H. K. O. visionary immediatly saw one SC application: magnets. Generation of large magnetic flux densities with little power => Project 10 T magnet in 1913.

Lead (Pb) magnet (1913)
9968 magnet



Tin (Sn) magnet (1913)
9969 magnet

Low magnetic field destroy superconductivity

⇒ Have to wait type II superconductors (end of 50')

(discovered in Soviet Union earlier - Shubnikov /Abrikosov)

# Introduction - History

- 1908 : Helium liquefaction
- 1911 : superconductivity discovery
- 1933 : Meissner effect
- 1957 : microscopic theory BCS
- 1958 : type II superconductors (NbSn, NbZr)

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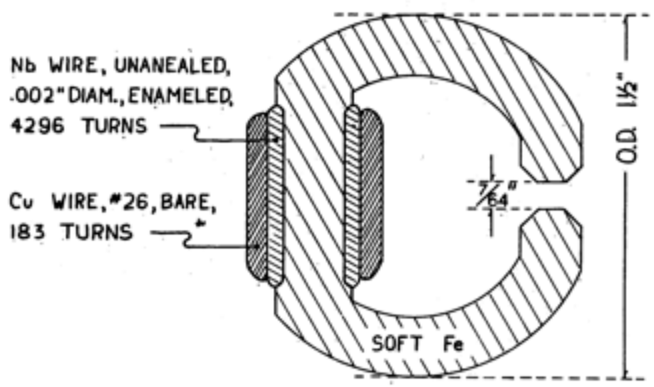
- 1960 : High current densities under fields for NbSn
- 1962 : first superconducting magnet

# First SC (NbZr) magnet

Oxford Instrument

Sir Martin Wood

March 1962.




G. Yntema, 1954

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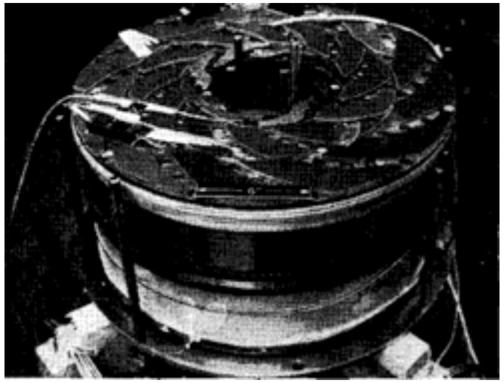
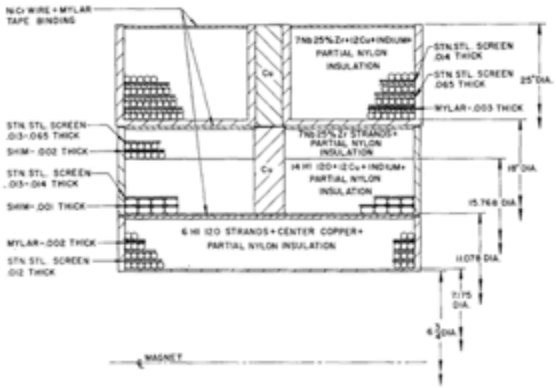
- 1963 : Quantum effects predicted by B. Josephson

- 1964 : First large scale application (Argonne bubble chamber)

# SC and high energy physics





**Superconductivity and Accelerators: the Good Companions**

Martin N. Wilson
Oxford Instruments, Tubney Woods, Abingdon, OX13 5QX, UK.


Roy Aleksan, ASC 2012

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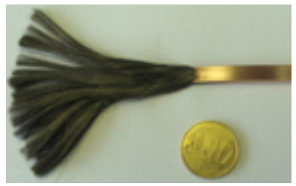


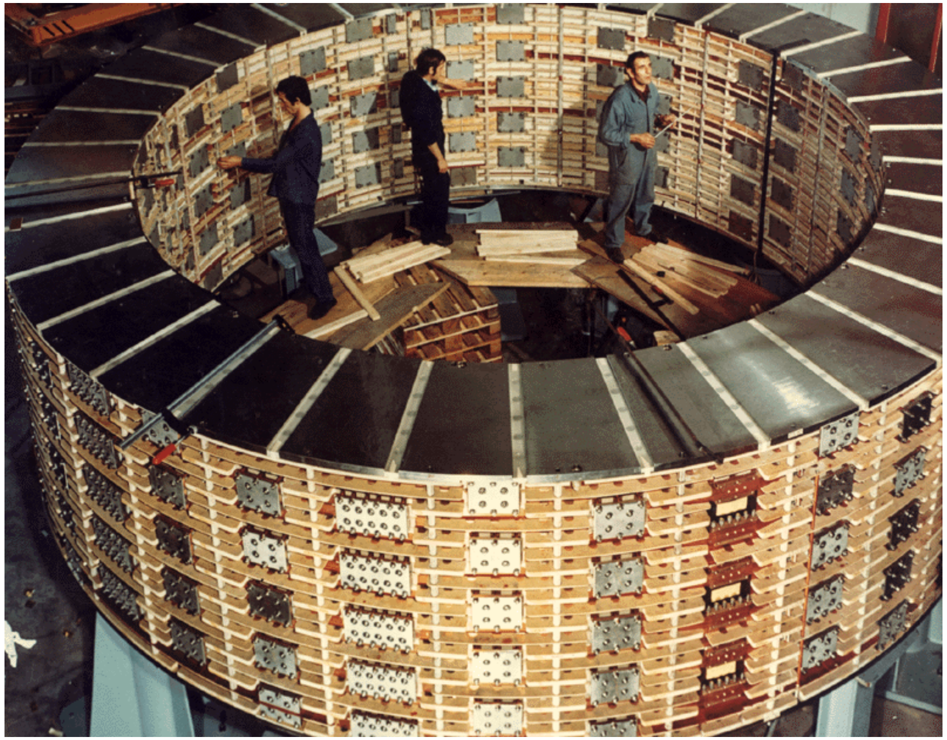
- 1974 : CERN bubble chamber (830 MJ)

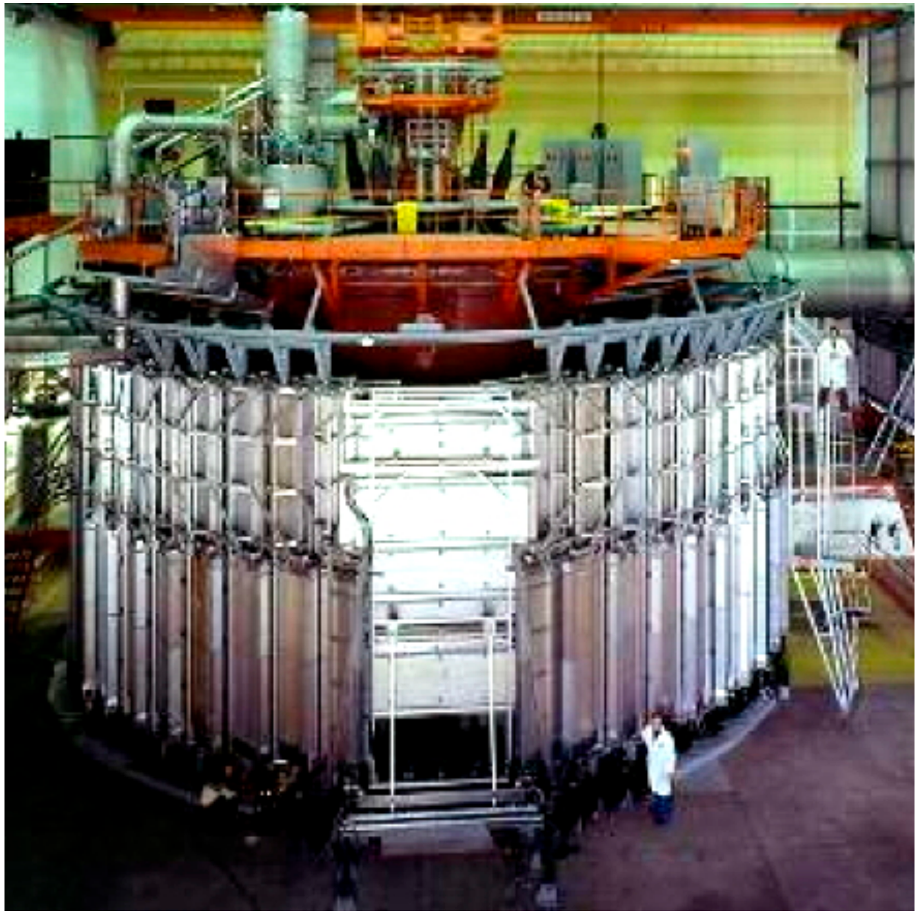
# CERN bubble chamber (BEBC)

$\varnothing_{int} = 3.7$ m

$B_o = 3.5$ T

$W_{mag} = 800$ MJ





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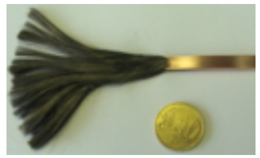

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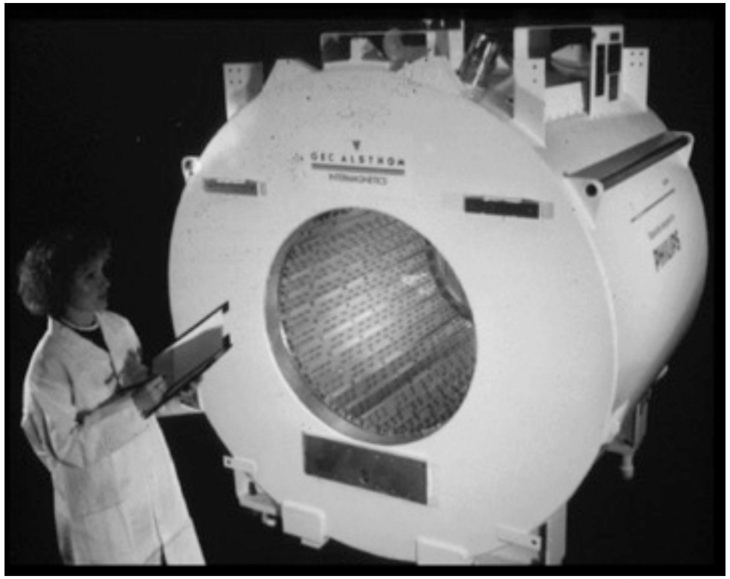
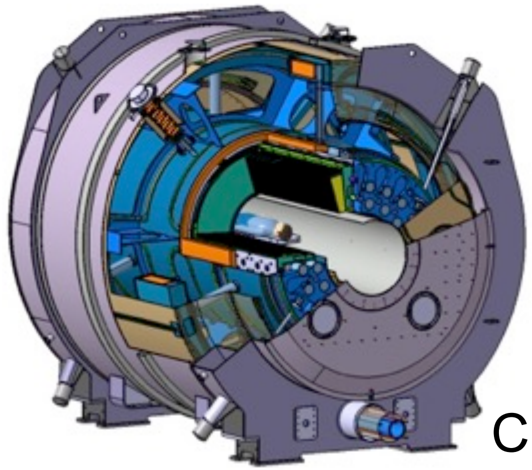
- 1982 : first MRI images (first commercial SC application)

# SC magnets and imaging system







CEA

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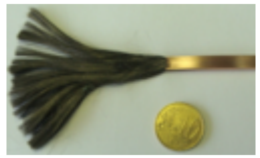
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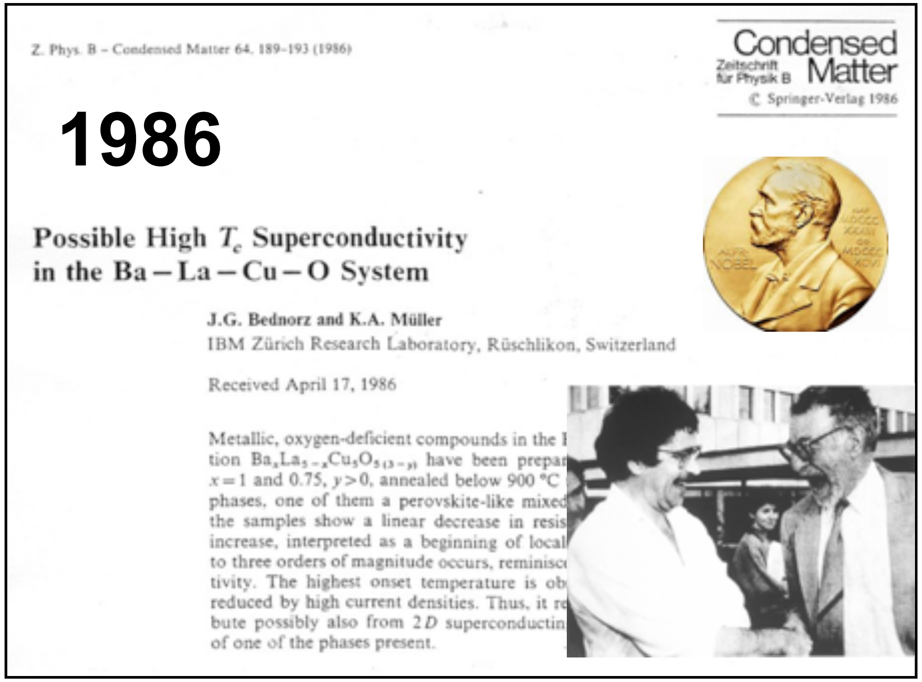
- 1987 : “high Tc” discovery

# High $T_c$ discovery


Z. Phys. B – Condensed Matter 64, 189–193 (1986)

Condensed Matter
Zeitschrift für Physik B
© Springer-Verlag 1986

**1986**

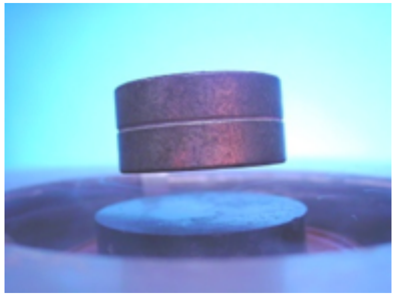
Possible High $T_c$ Superconductivity
in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

Metallic, oxygen-deficient compounds in the I
tion $Ba_xLa_{5-x}Cu_5O_{5(3-y)}$ have been prepar
$x=1$ and 0.75, $y>0$, annealed below 900 °C
phases, one of them a perovskite-like mixed
the samples show a linear decrease in resis
increase, interpreted as a beginning of local
to three orders of magnitude occurs, reminisc
tivity. The highest onset temperature is ob
reduced by high current densities. Thus, it re
bute possibly also from 2D superconductin
of one of the phases present.

**1987**



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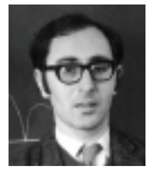

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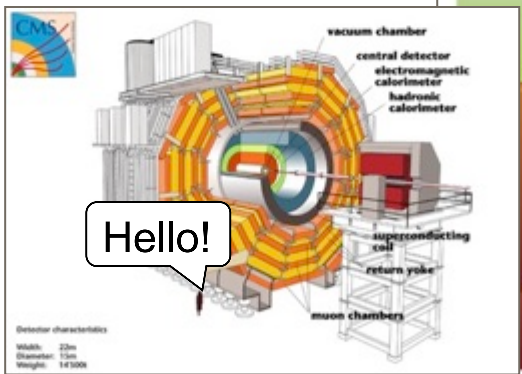
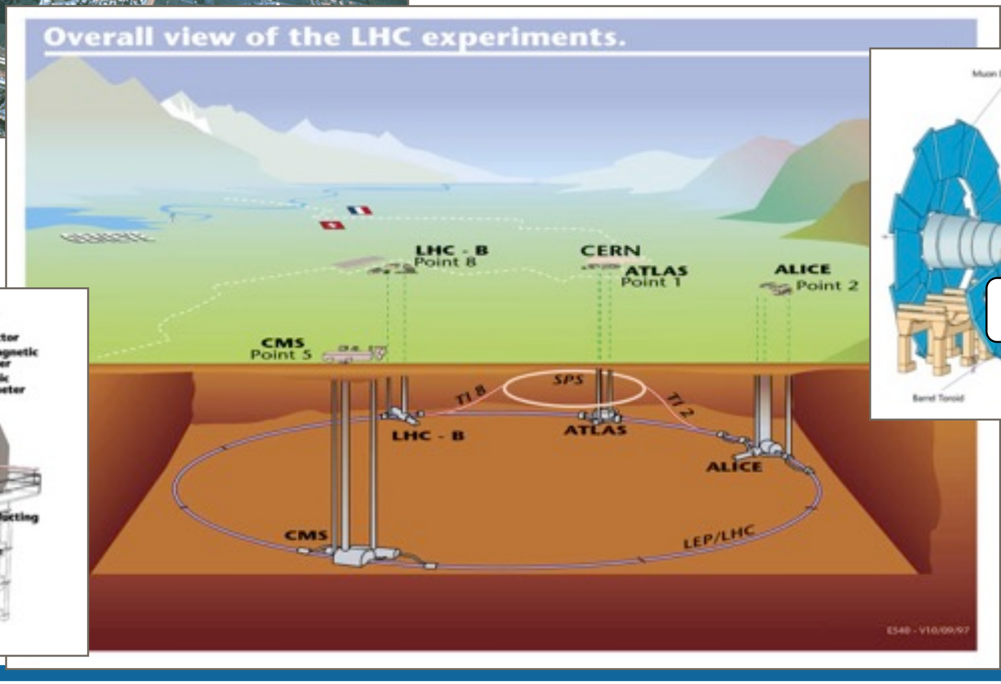
- 1987 : “high Tc” discovery
- 2007 : starting of the biggest SC system: LHC

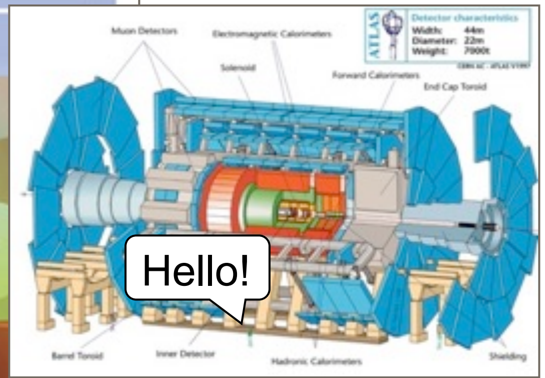
# LHC: bigest SC system

1232 dipoles
392 main quadrupoles
+ experiences (ATLAS, CMS…)

CMS

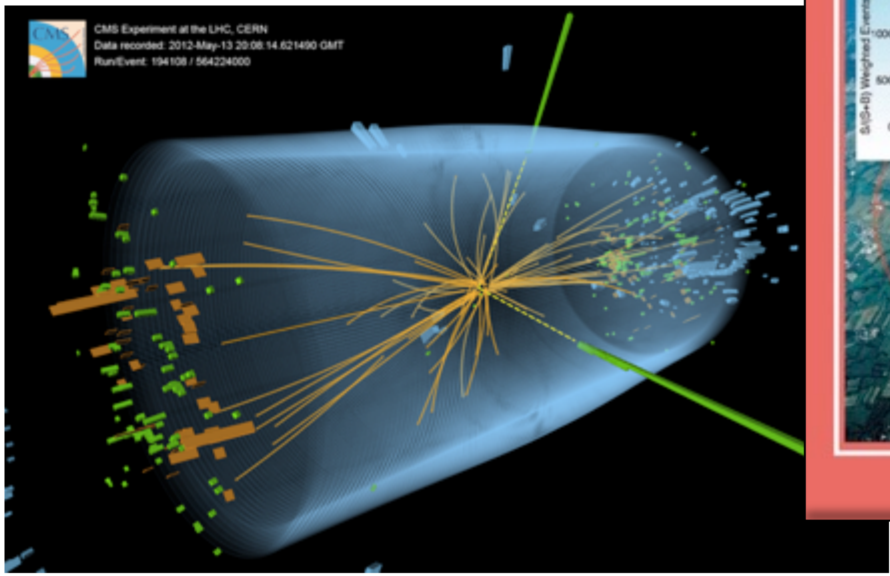


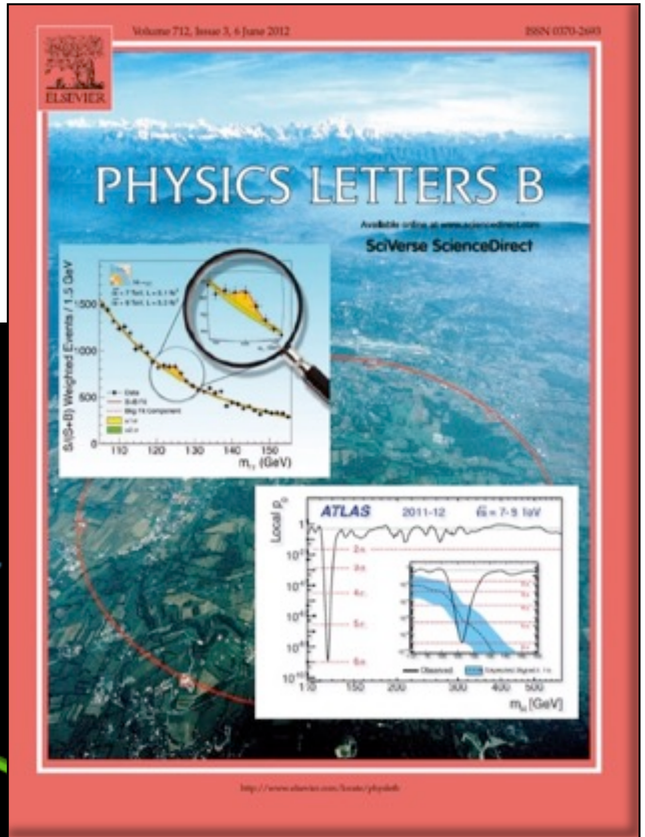




ATLAS

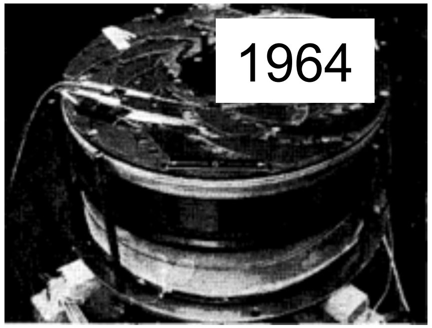
# July 4, 2012: day H at CERN


# Summary: applied superconductivity

End of 50': high $J_c$ under field
Applied SC is born


Argonne magnet



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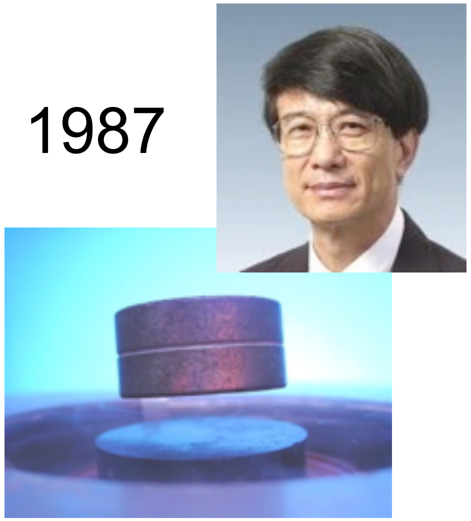
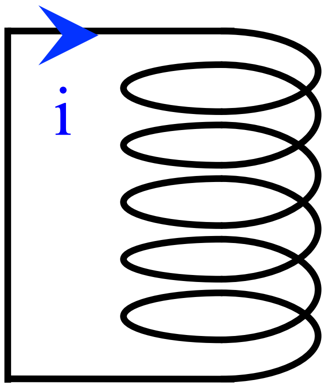
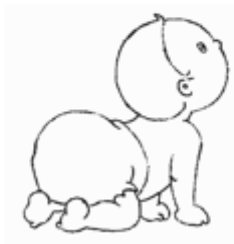


# Superconductivity for large scale applications

# Superconductor:
# **perfect conductor, but not ideal!**

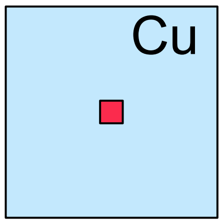
$P_{DC} = 0$



100.00 A

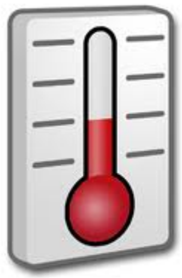
90.00 A
99.93 A




SC
Hundred
$A/mm^2$

$T < T_c$

$\Rightarrow$ Cryogenic

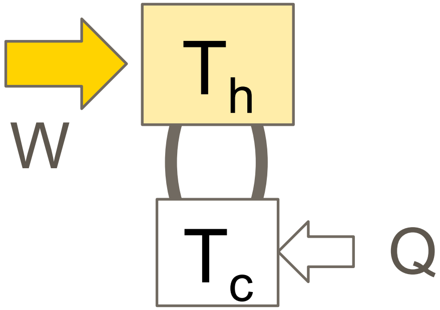


- Cost
- ***Psychological Fear***

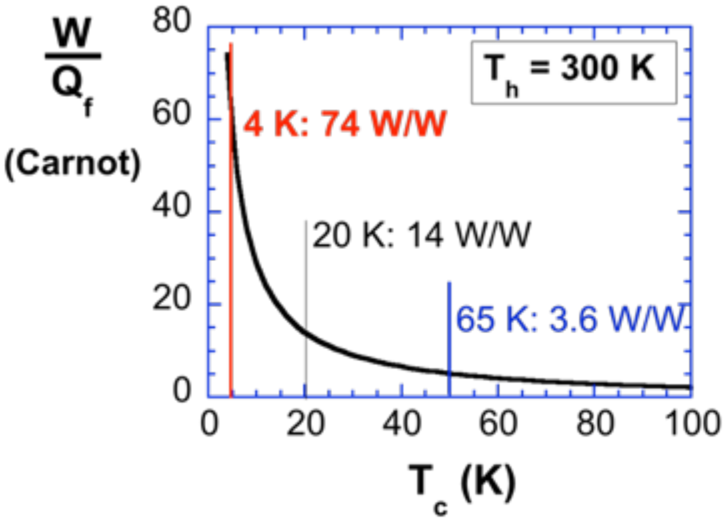
$+ H < H^* \;\&\; J < J_c$

+ Superconductor cost

# Cost of energy removal at low temperatures

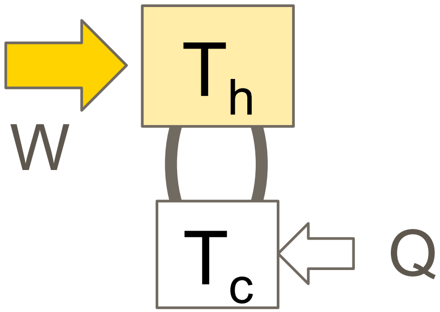



$$\frac{W_{min}}{Q} = \frac{T_h - T_c}{T_c} \quad \frac{1}{\eta_{ref}}$$




- 1 liter He (LTS) ≈ 5 €
- 1 liter $N_2$ (HTS) ≈ 0.08 €

# Cost of energy removal at low temperatures

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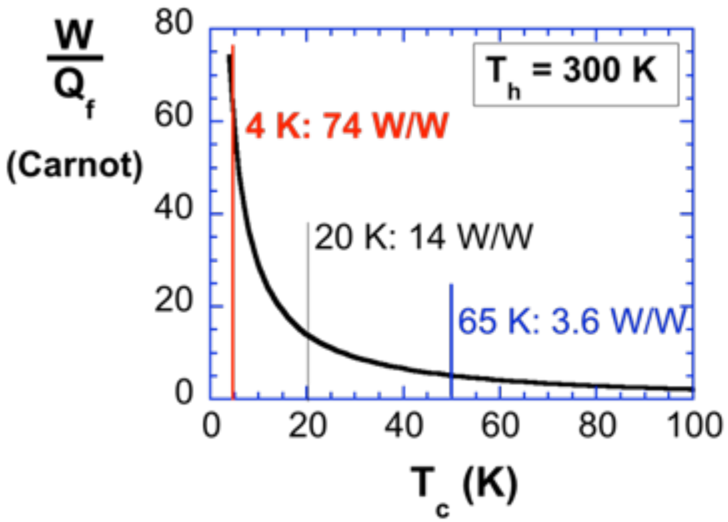


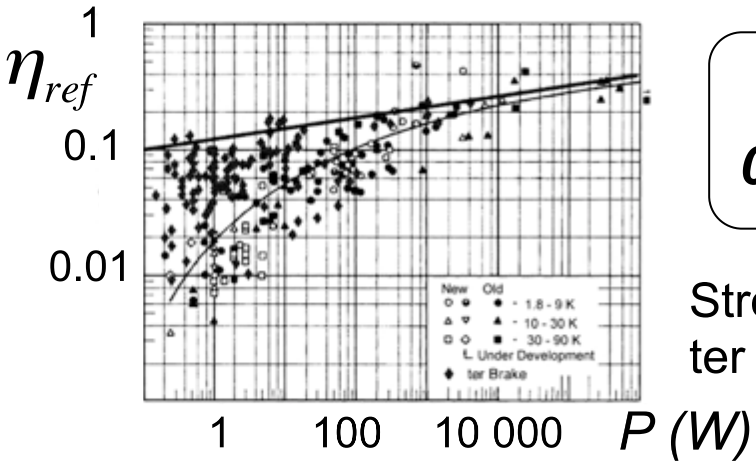


$\eta_{ref}$

***0.1 – 0.3***

Strobridge
ter Brake

- 1 liter He (LTS) ≈ 5 €
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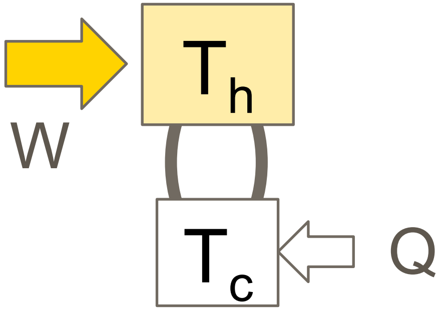
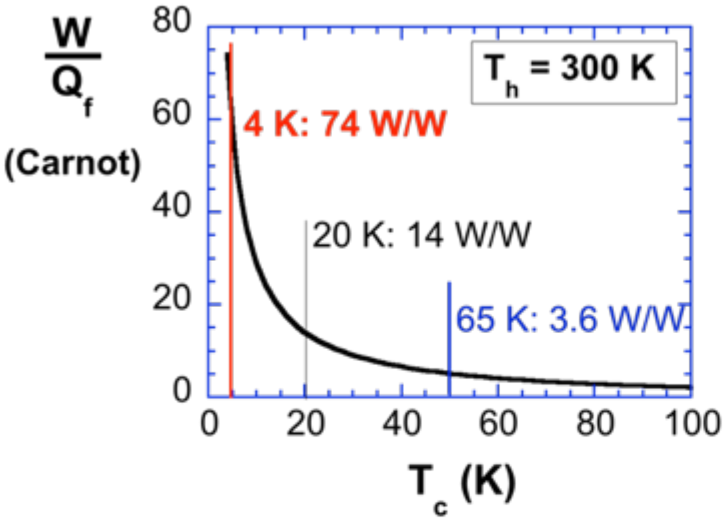
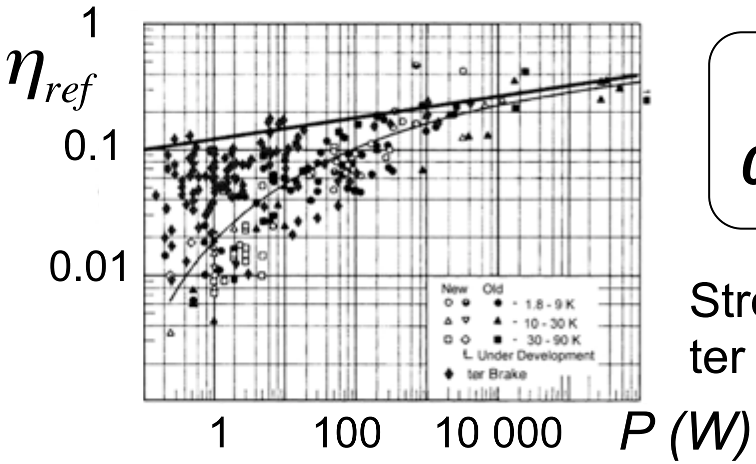
# Cost of energy removal at low temperatures




$$\frac{W_{\min}}{Q} = \frac{T_h - T_c}{T_c} \frac{1}{\eta_{ref}}$$

**=> Strong interest to operate at "high" temp.**




$\eta_{ref}$

***0.1 – 0.3***

Strobridge
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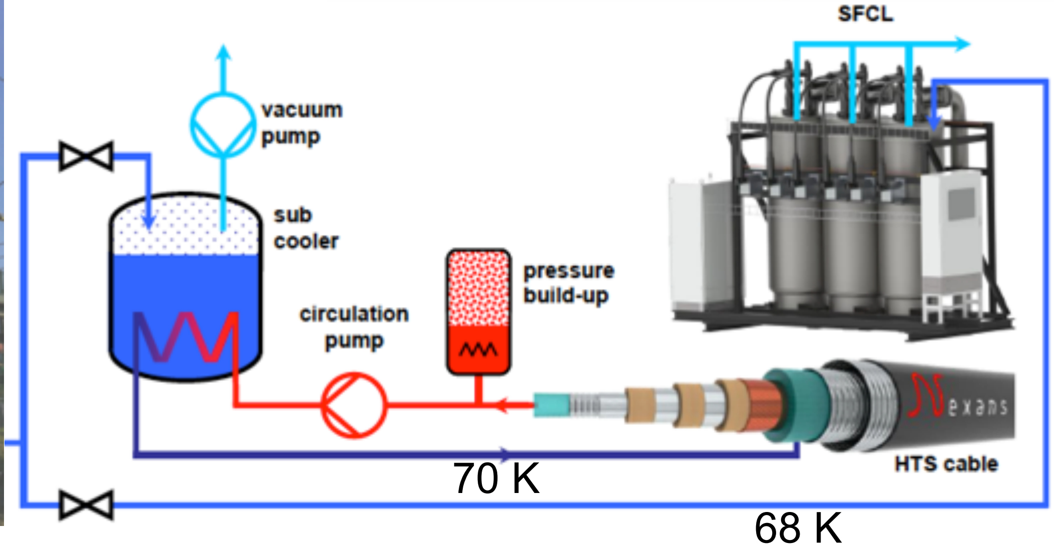
- 1 liter $N_2$ (HTS) ≈ 0.08 €

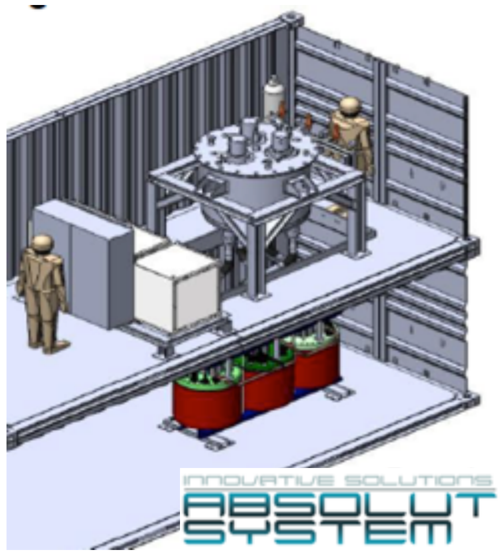
Liquid $N_2$: environ$^t$ friendly cooling fluid & electrical insulation

# Two main cooling solutions

Comonly used in industry


LN2 production overstep 60 000 tons/day for some large plants




Liquid $N_2$: environ$^t$ friendly cooling fluid & good electrical insulater

# AC losses

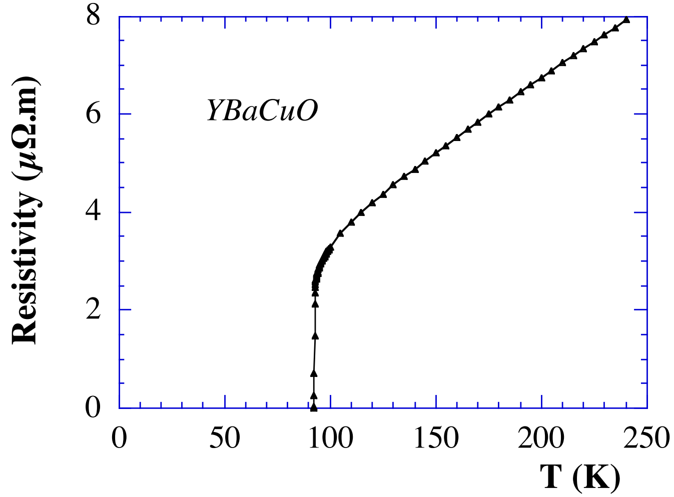
$$P_{DC} = 0 \text{ But } P_{AC} \neq 0$$

Faraday law: $\overrightarrow{curl}\,\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$ $\qquad \vec{E} + \vec{J} : losses \left( \delta P = \vec{E}\,\vec{J} \left[ W / m^3 \right] \right)$

AC losses may be reduced by a suitable structure but not suppressed.

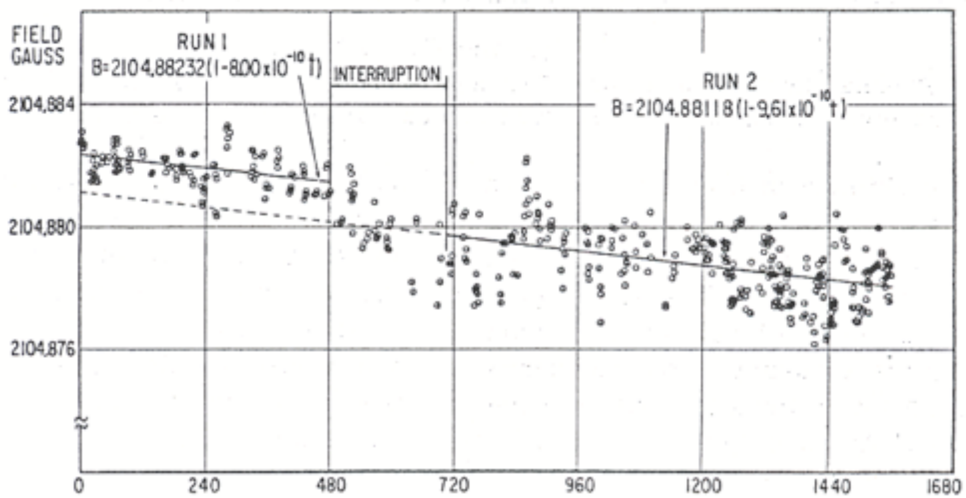
**=> Interest more in DC than AC**

# Perfect conductivity $P_{DC} = 0$

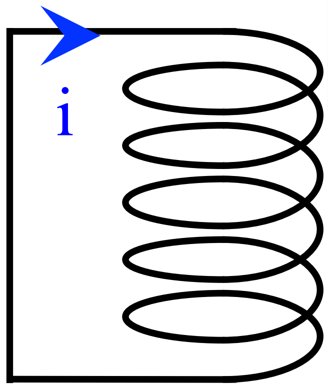



Field decrease measurement => $\tau$

Mills & Files experiment (1962)




(hours)




100.00 A

90.00 A
99.93 A

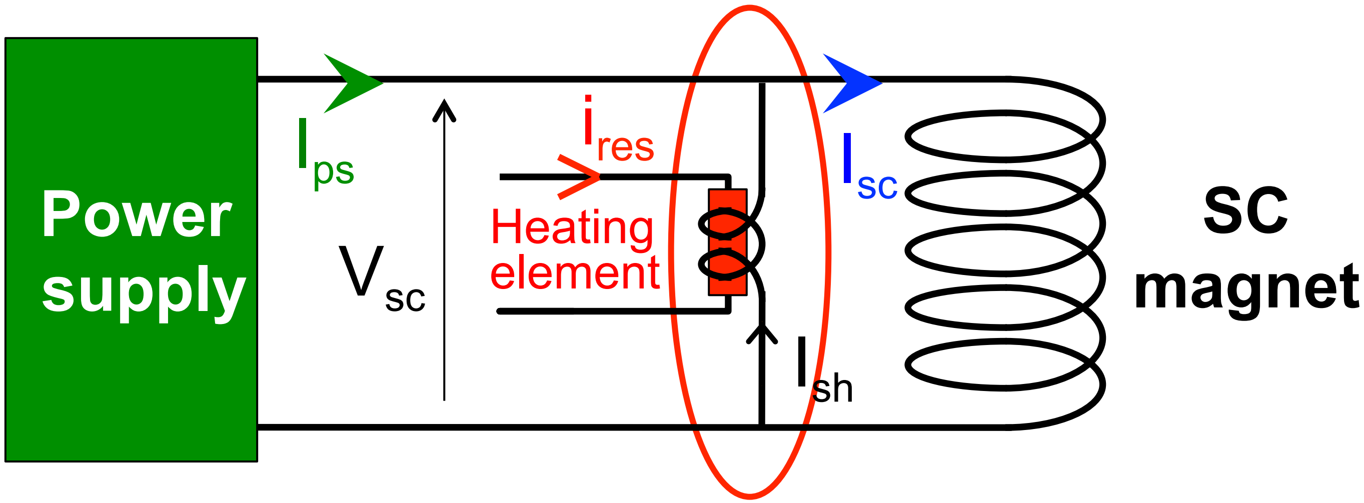
*Experimental time constants:*

Test 1 : $\tau$ = 144 000 years

=> $\rho < 10^{-25}\ \Omega$m

# Persistent mode

How to induce a current in a SC coil?

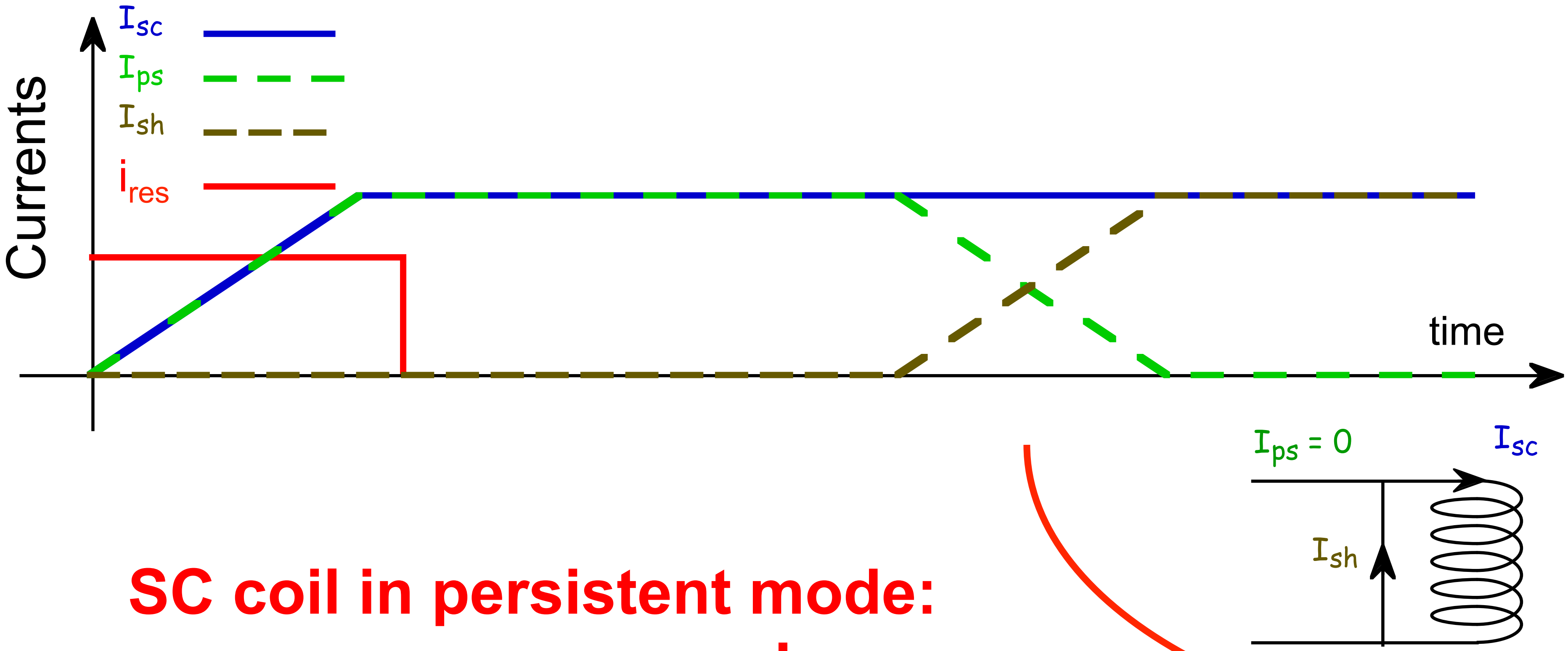



**SC shunt with thermal control**

- **heating: normal state, high resistance**
- **no heating: SC state**

**(SC shunt with field control as well)**

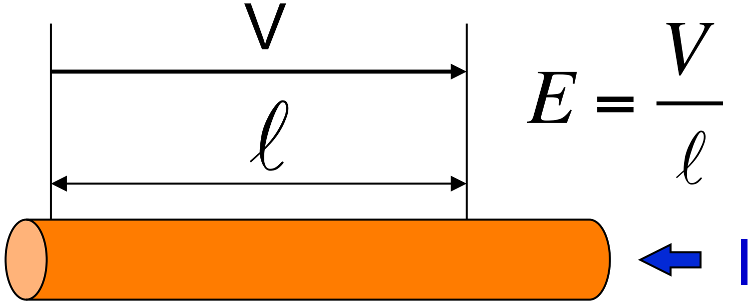
# Current supply: end of cycle




**SC coil in persistent mode:**
**- no power supply**
**- ultra-stable current**

# Perfect conductivity

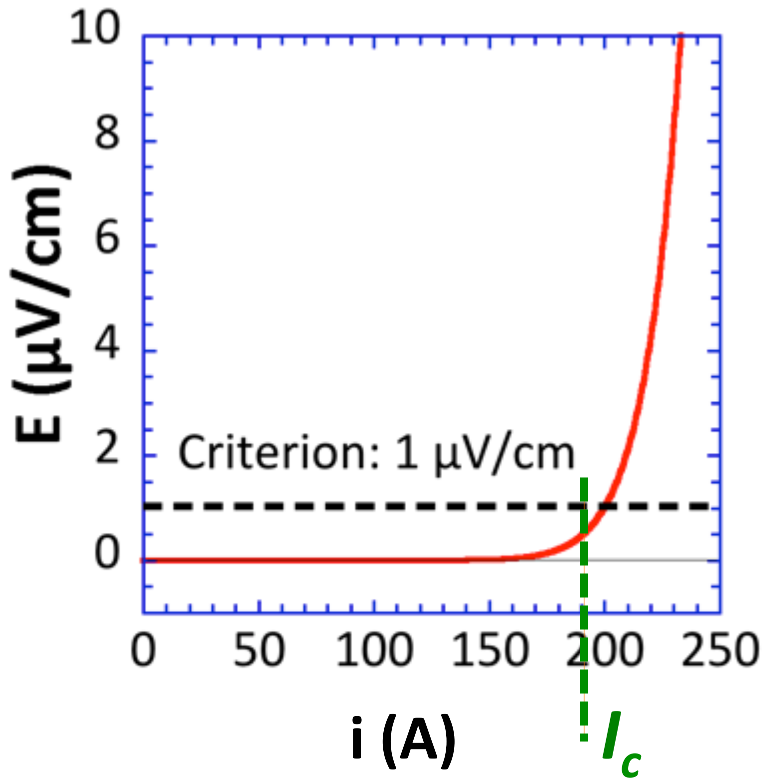
$\rho = 0$ => I/J infinite?




$$E = \frac{V}{\ell}$$

V: voltage (V)

E: electric field (V/m) (µV/cm)



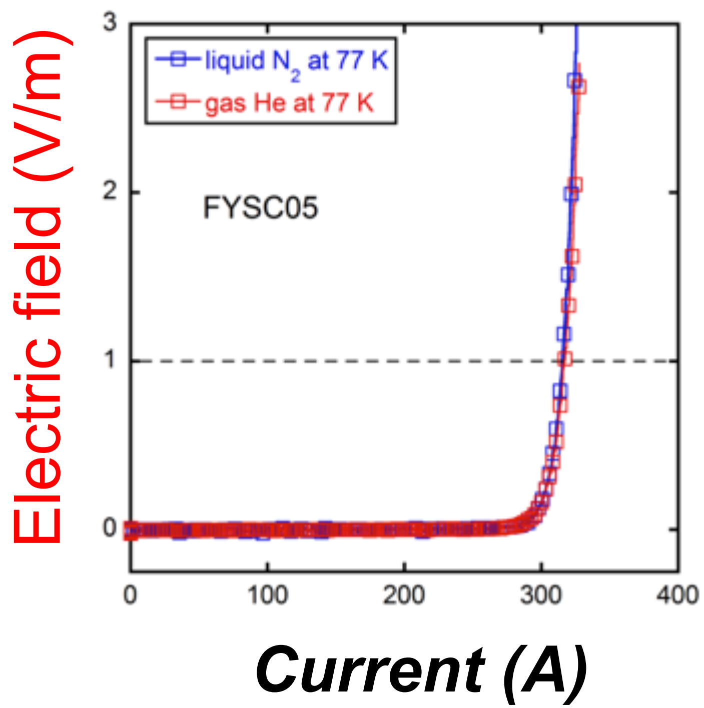

Criterion to define $I_c$

E = 100 µV/m (1 µV/cm)

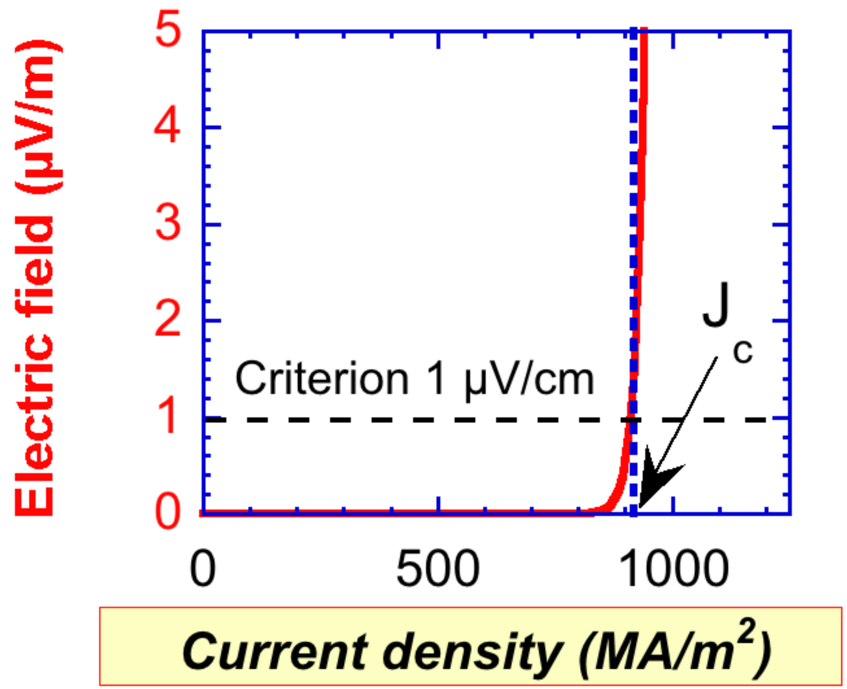
International standart

# E-J Characteristics

Experimental E(I) curve

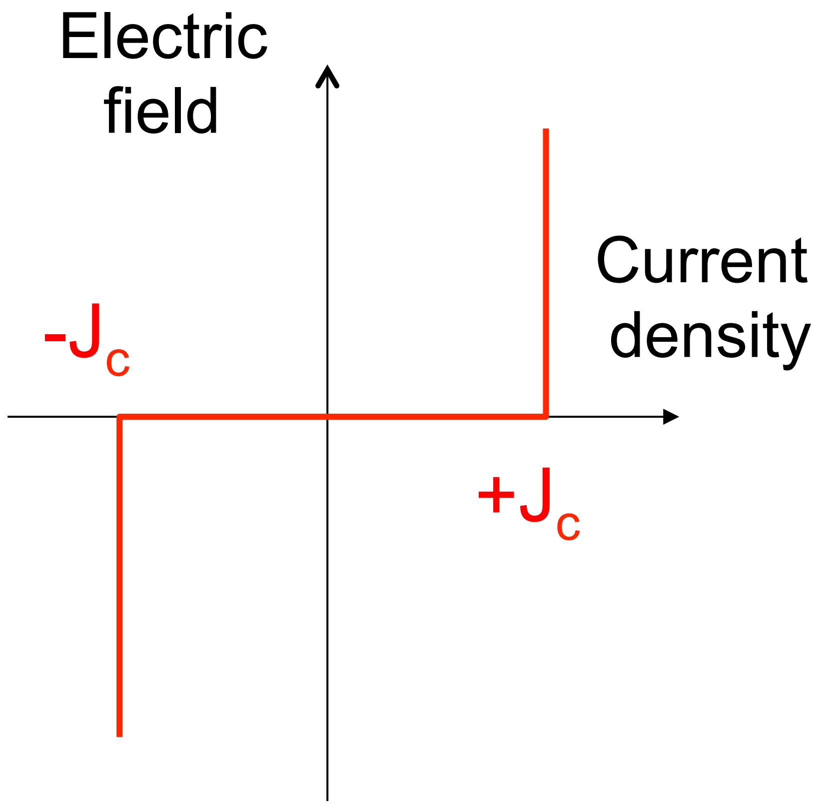



MODEL




- Relation valid for local current density
- Infinite stiffness

# E-J – Critical State Model




Local current density

ONLY 3 values

- 0
- $+ J_c$
- $- J_c$

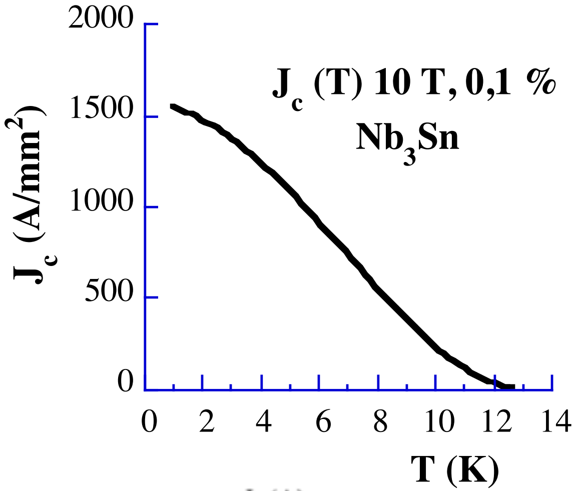
Critical state model (CSM)

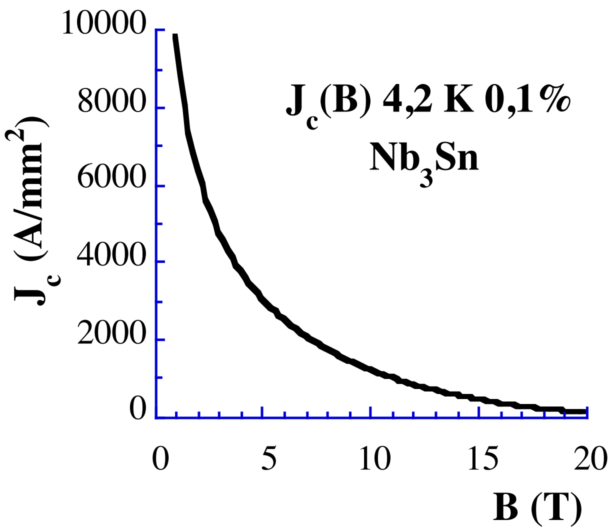
Bean model:
Critical state model
+
$J_c$ constant (not function of B)

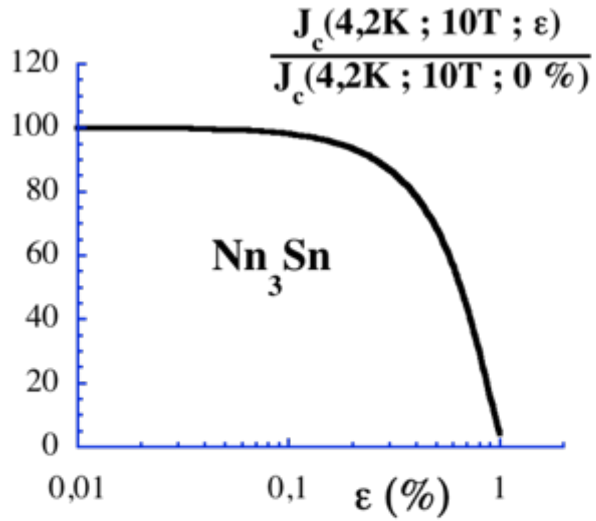
Bean Model very used
Especially for analytical expressions

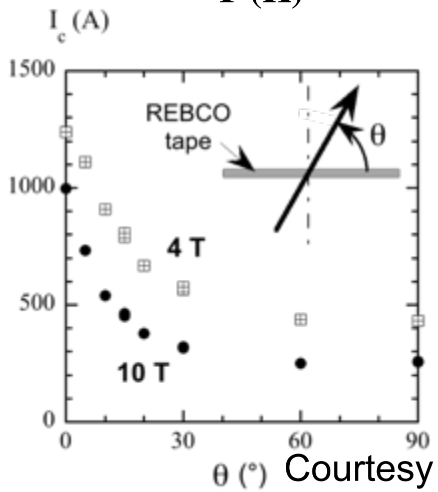
# Critical current density $J_c$

$$J_c = J_c\,(T, B, (\theta), (\varepsilon))$$

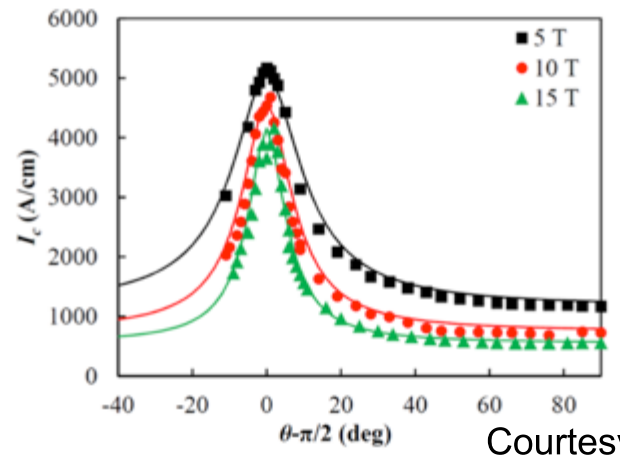


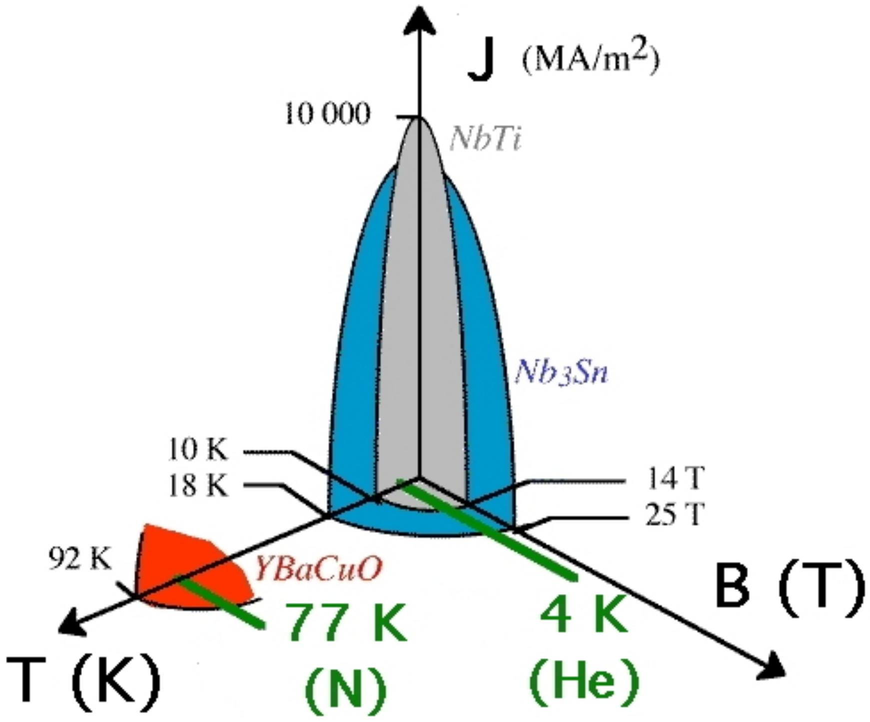

Courtesy T. Benkel

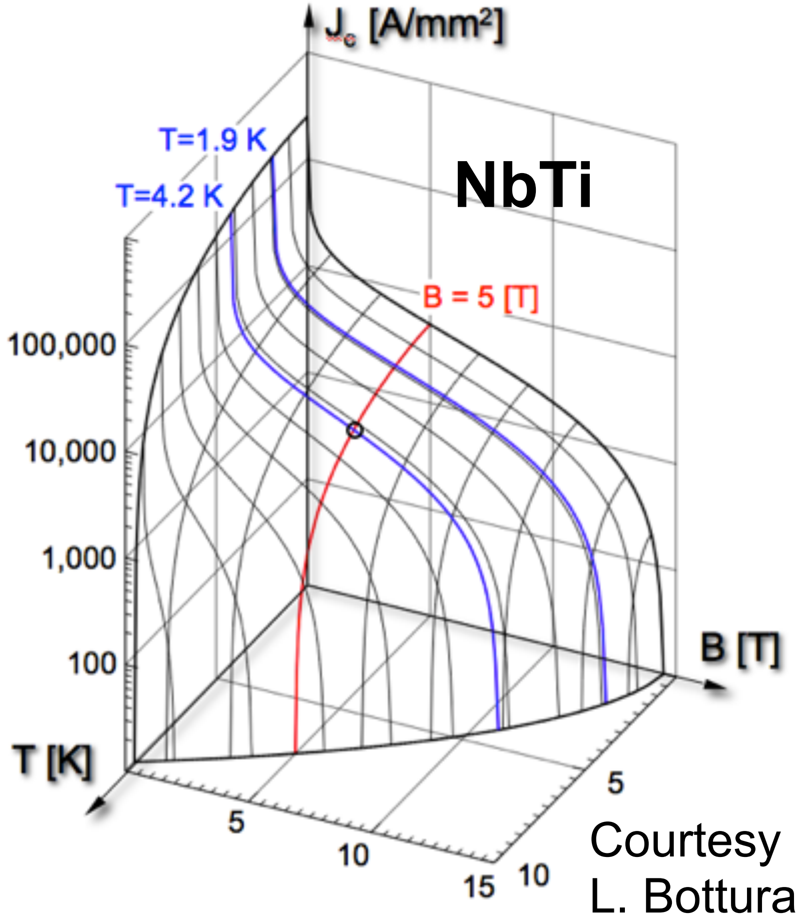



Courtesy J. Fleiter

# Critical surface

- Three main dependent limits ($T_c$, $B_c$, $J_c$): critical surface



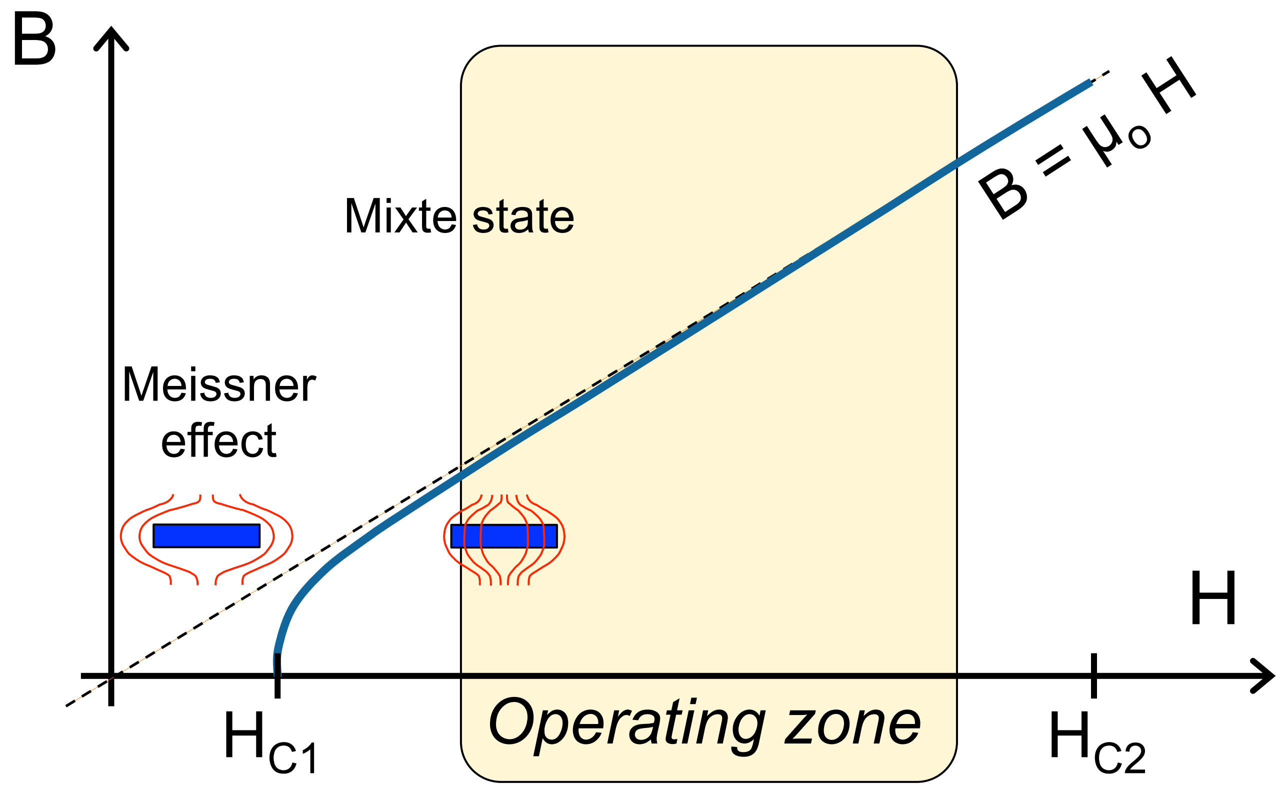

Courtesy L. Bottura

- $T_c$ & $H_c$: intrinsic
- $J_c$: elaboration (pinning)

# B-H characteristic

B = 0 in a superconductor (Meissner effect)

*Forget the Meissner effect for large scale applications!*




$$\boxed{B \approx \mu_o H}$$

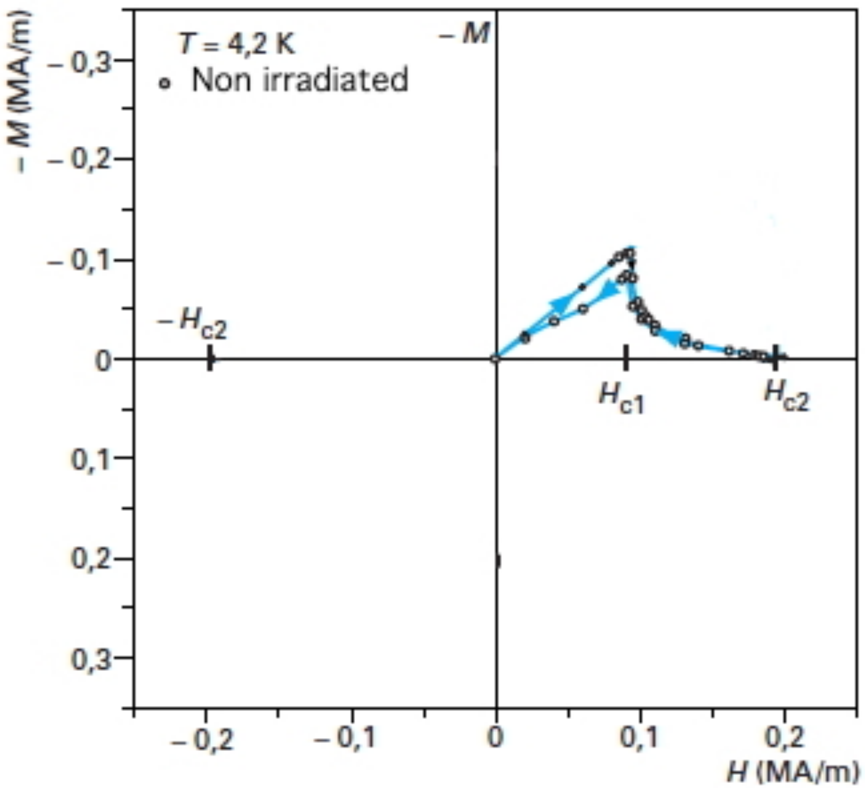
$\mu_o H_{C1}$ (0 K)

| |
|---|
| NbTi: 10 mT |
| $Nb_3Sn$: 17 mT |
| YBCO: ≤ 70 mT |

***Not to scale!***

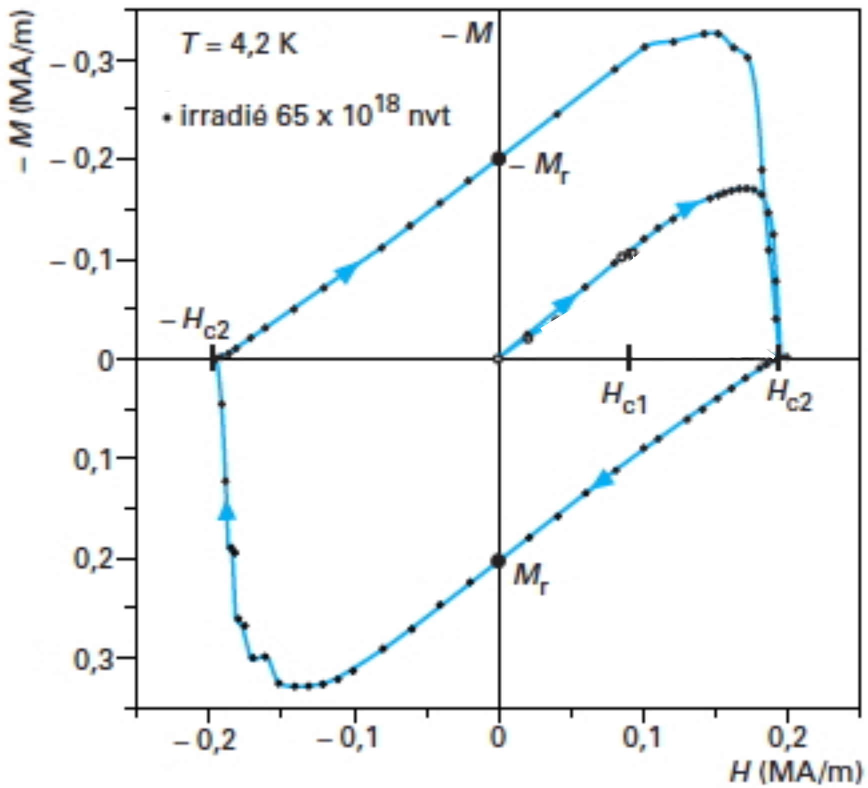
# B-H characteristic (M-H)

$$B = \mu_o (H + M) \quad \Rightarrow \quad -M = H - \frac{B}{\mu_o}$$



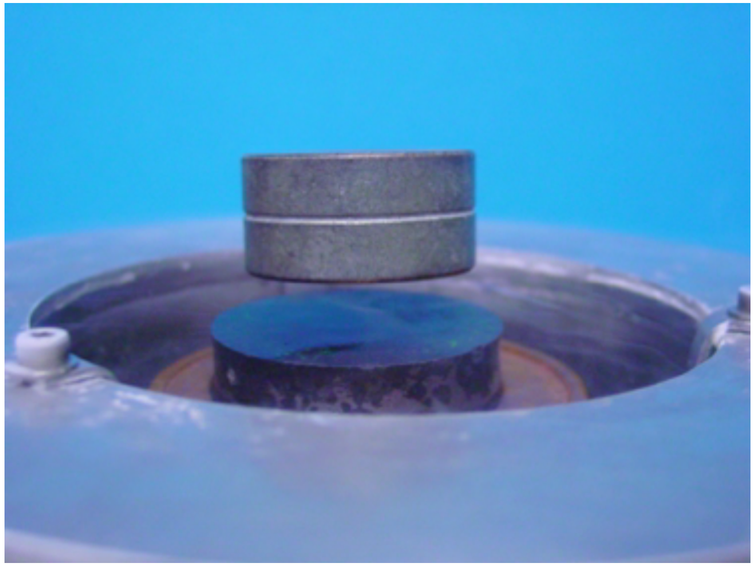

Non irradiated Nb sample

From Sekula and Kernohan

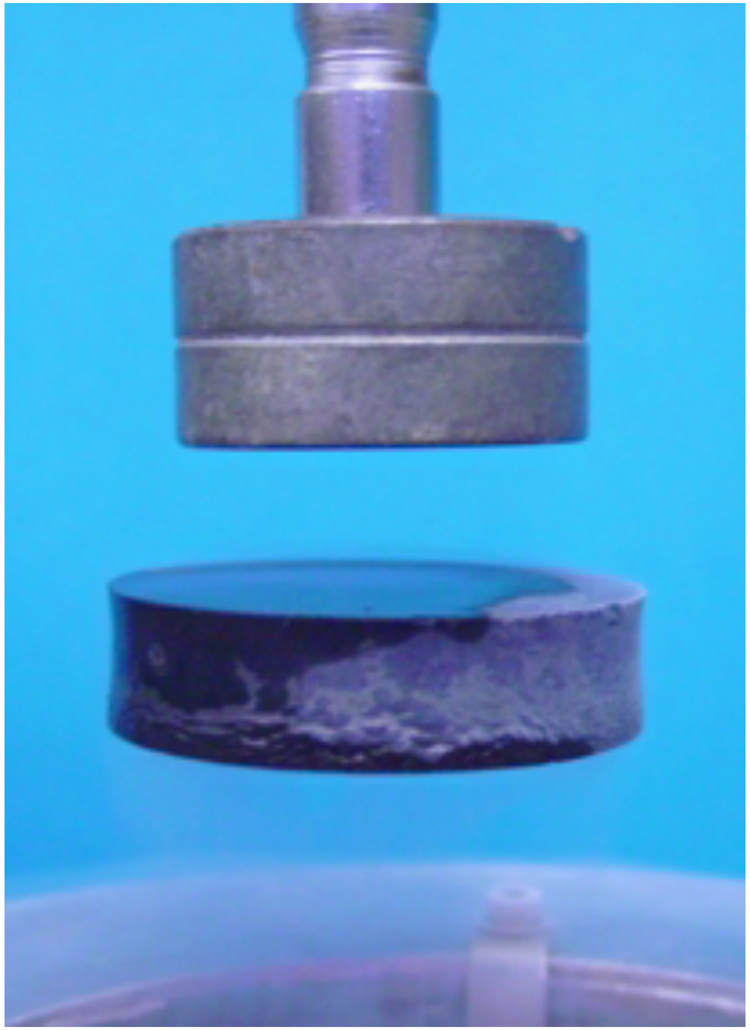



Irradiated Nb sample

# Levitation experiment



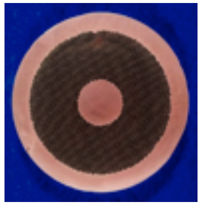
Based on induced current
(Lenz law)
and absence of damping



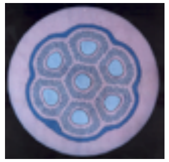
# Superconducting wires

**Low $T_c$**

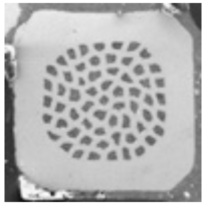
- NbTi (3000 t/y) – *9.5 K*



- $Nb_3Sn$ (25 t/y ~~ITER~~) – *18 K*

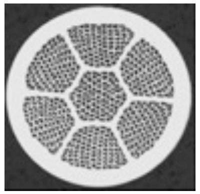


- $MgB_2$ – *39 K*

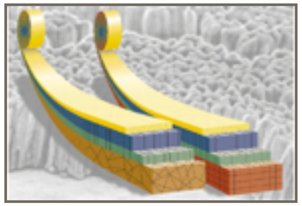


**High $T_c$**

- BiSrCaCuO (1G)
  *85 – 110 K*



- ReBaCuO (2G) – *90 K*



**Thin tape**

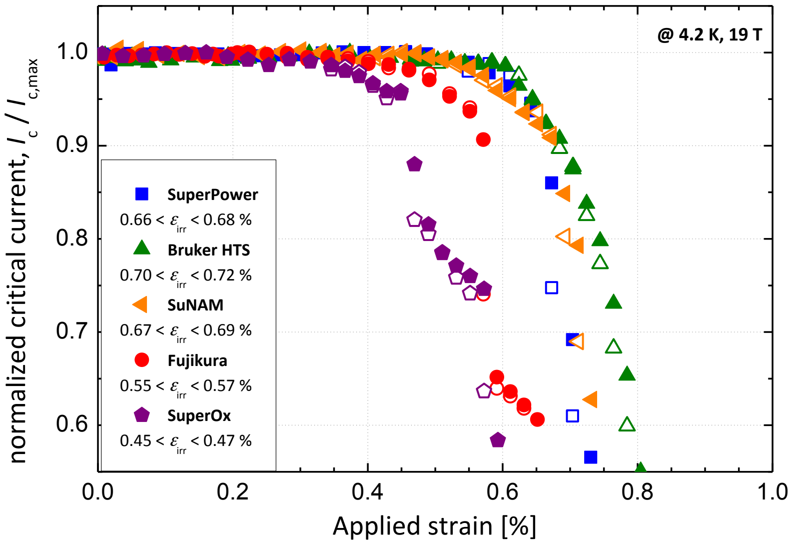
# Mechanics: must be stress tolerant!

Fundamental for LTS magnet

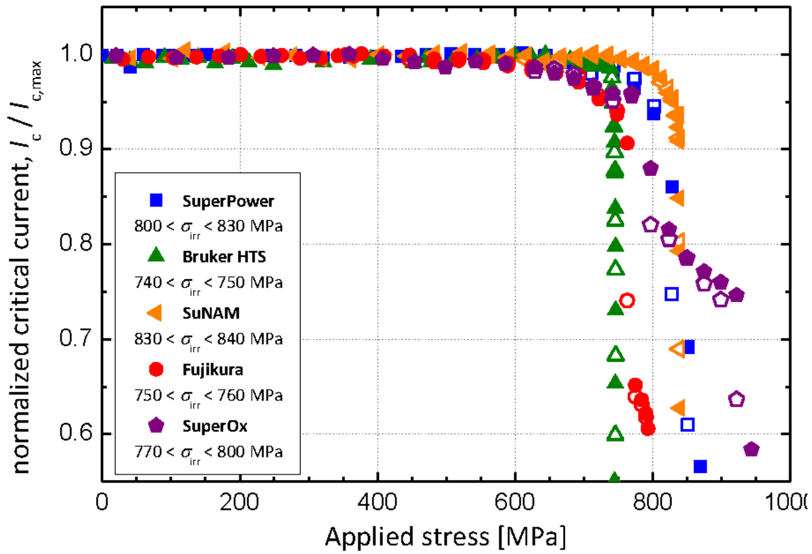
- Ultra sensitive to mechanical perturbations

High mechanical performances often required

- Magnets: high J and B => high Lorentz force (JB)
- Solenoid: $\sigma$ = J B R (independant turns)




*$I_c$ vs. axial stress*



# Summary

- Superconductivity market exists
  - 5 185 Meuros in 2012
  - Dominated (> 60 %) by MRI and NMR
  - Fully still dominated by LTS (> 90 %), NbTi
- Persistent mode for SC magnet
  - Resistive (inductive) shunt to inject current
  - Very stable current and no power supply

# Outline

- Introduction
- Historical references
- Superconductivity for large scale applications
- Superconducting material (REBCO tapes)
- Superconducting applications
- Fault Current Limiter (FCL)

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# Superconductivity II
## Superconducting Materials - ReBCO tapes


Pascal Tixador

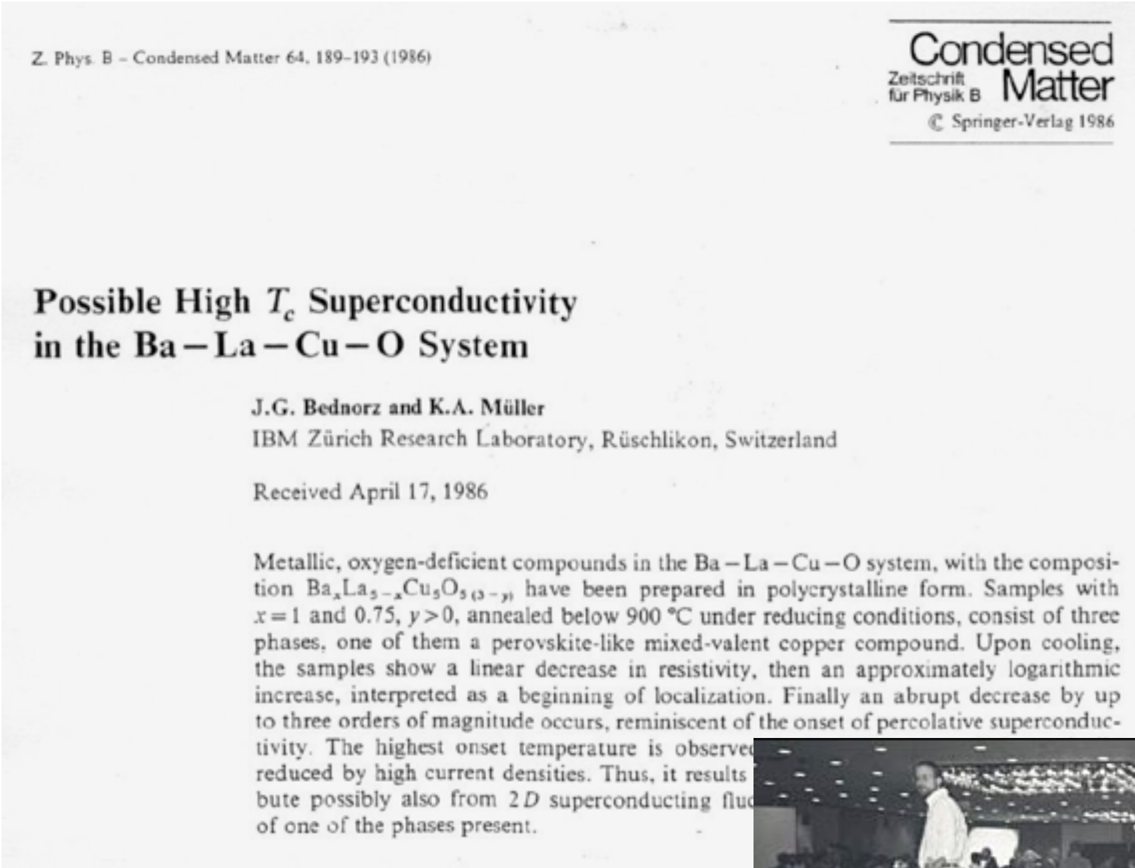
Grenoble-INP, Université de Grenoble Alpes, CNRS
G2ELab, Institut Néel


UNIVERSITÉ Grenoble Alpes · Grenoble INP · CNRS · NÉEL institut · G2E Lab

# Summary LTS

- LTS by far the most used SC
  - 5050 M€ / 30 M€ for HTS (SC devices 2011)
- Two main LTS materials: NbTi & $Nb_3Sn$
- NbTi is by far the most used SC
  - 3000 tons/year
  - Easy to use and handle
  - Limited in field: 9 T (4 K) & 12 T (1.8 K)
- $Nb_3Sn$ complex to handle (brittle, reaction)
  - B < 23.5 T (1.8 K)

# HTS breakthrough


Z. Phys. B – Condensed Matter 64, 189–193 (1986)

Condensed Matter
Zeitschrift für Physik B
© Springer-Verlag 1986

**Possible High $T_c$ Superconductivity in the Ba–La–Cu–O System**

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

Received April 17, 1986

Metallic, oxygen-deficient compounds in the Ba–La–Cu–O system, with the composition $Ba_xLa_{5-x}Cu_5O_{5(3-y)}$ have been prepared in polycrystalline form. Samples with $x=1$ and 0.75, $y>0$, annealed below 900 °C under reducing conditions, consist of three phases, one of them a perovskite-like mixed-valent copper compound. Upon cooling, the samples show a linear decrease in resistivity, then an approximately logarithmic increase, interpreted as a beginning of localization. Finally an abrupt decrease by up to three orders of magnitude occurs, reminiscent of the onset of percolative superconductivity. The highest onset temperature is observe[...] reduced by high current densities. Thus, it results [...] bute possibly also from 2D superconducting flu[...] of one of the phases present.

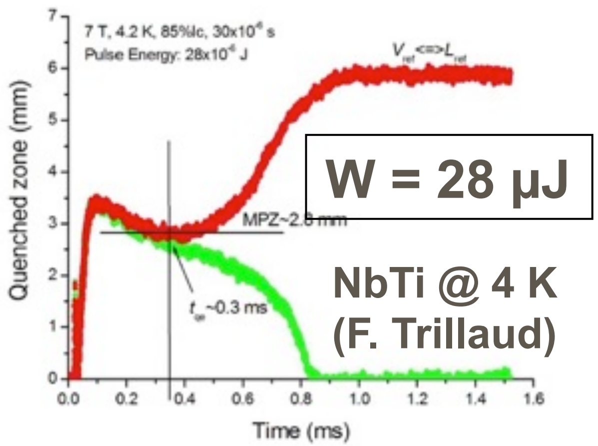
YBaCuO
January 1987

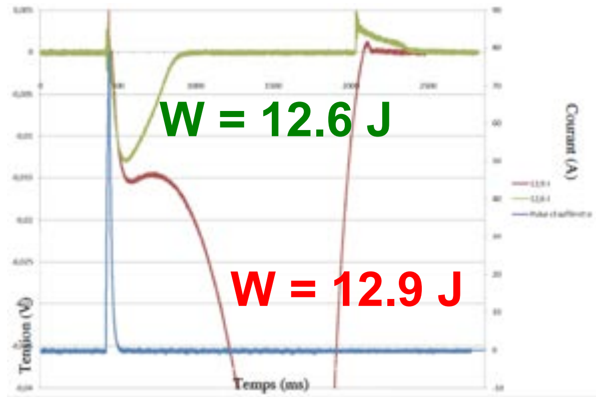
Woodstock of
Physics 1987

# HTS interest

## Stability

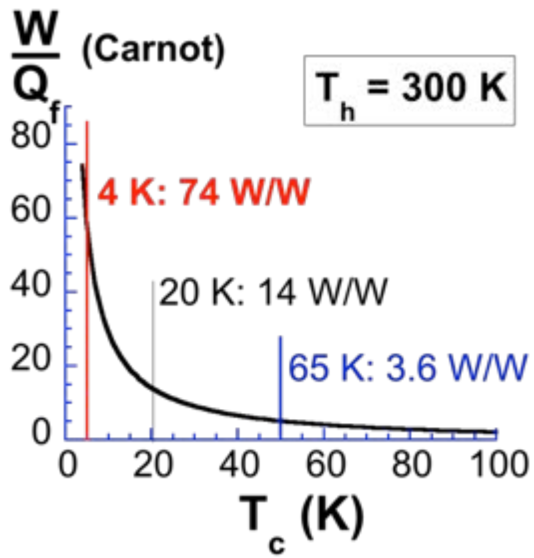
(behaviour against perturbations)




REBCO @ 4 K (B. Vincent)

## Cryogenics

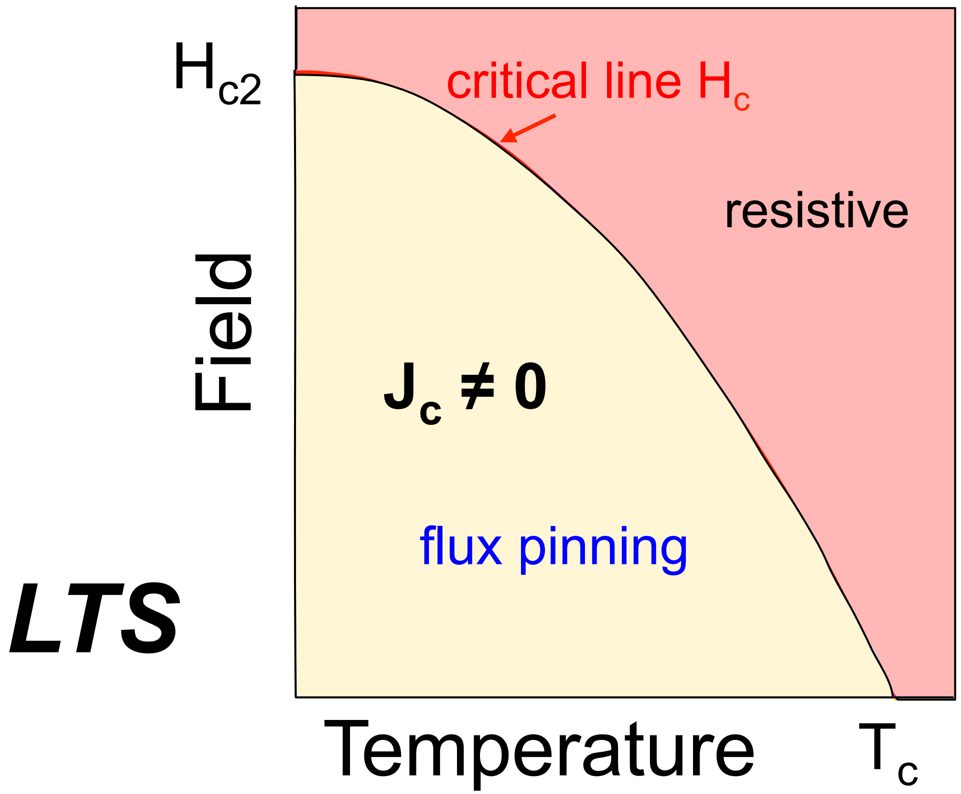



- 1 liter He: 5 €
- 1 liter $N_2$: < 0.1 €

Liquid $N_2$:
industrial fluid

# HTS specificities

Phase diagram of a superconductor - LTS

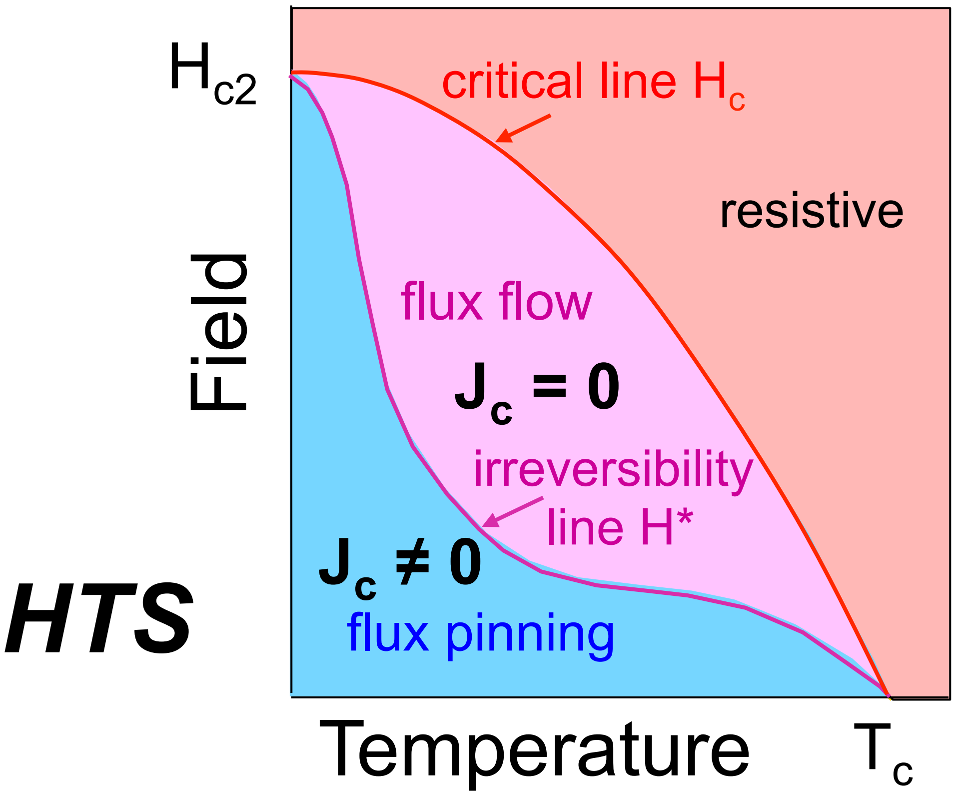



One single line

- Critical line $H_c(T)$
  - Relevant: $J_c = 0$

# HTS specificities

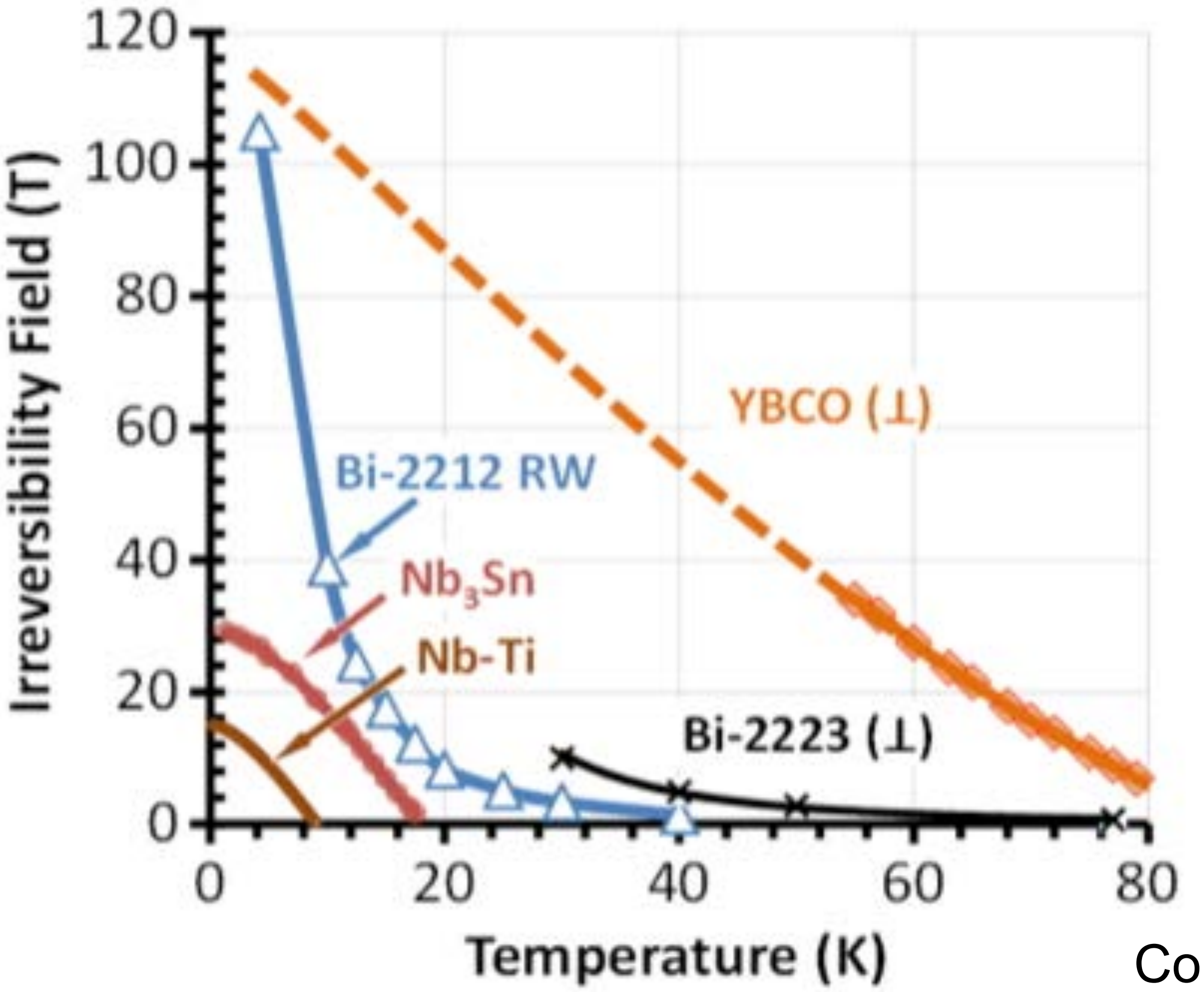
## Phase diagram of a superconductor - HTS




Two lines

- Critical line $H_c(T)$
  - Not relevant: $J_c = 0$
- Irreversibility line $H^*(T)$
  - Relevant: $J_c \neq 0$

# Irreversibility lines

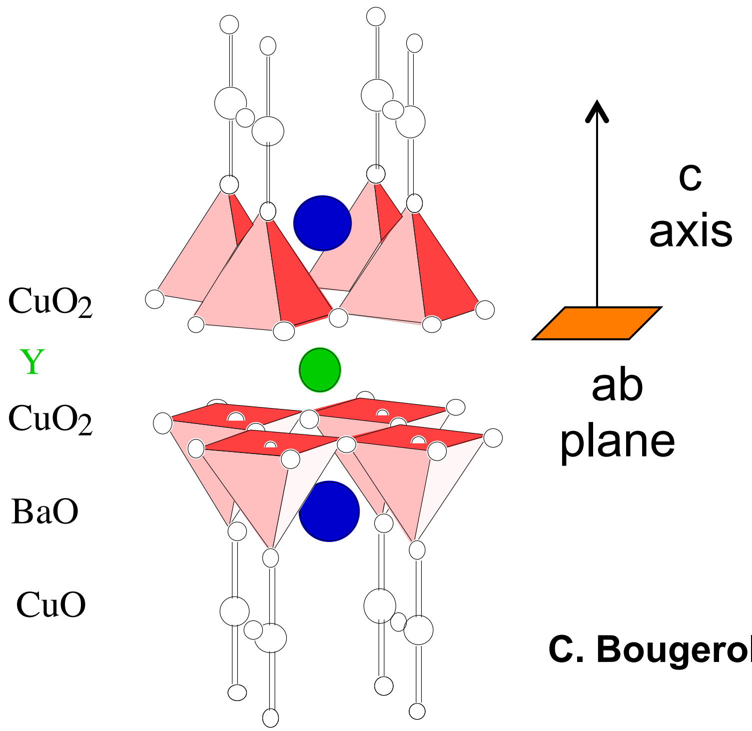



Strong interest for YBaCuO

Courtesy D. Larbalestier

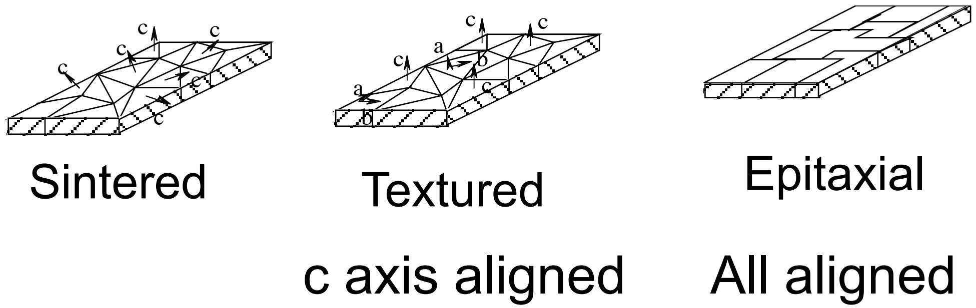
# HTS specificities

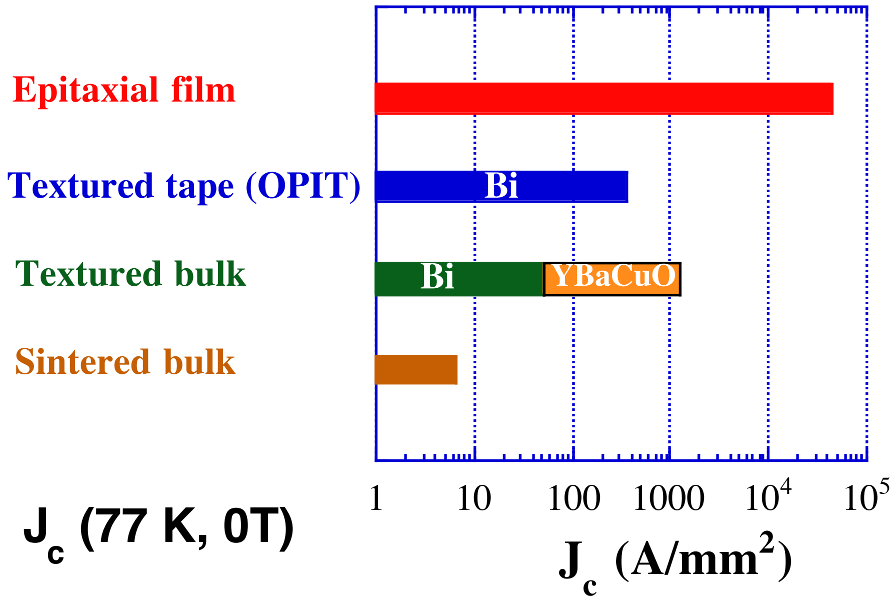
## High anisotropy




$YBa_2Cu_3O_7$

**C. Bougerol**



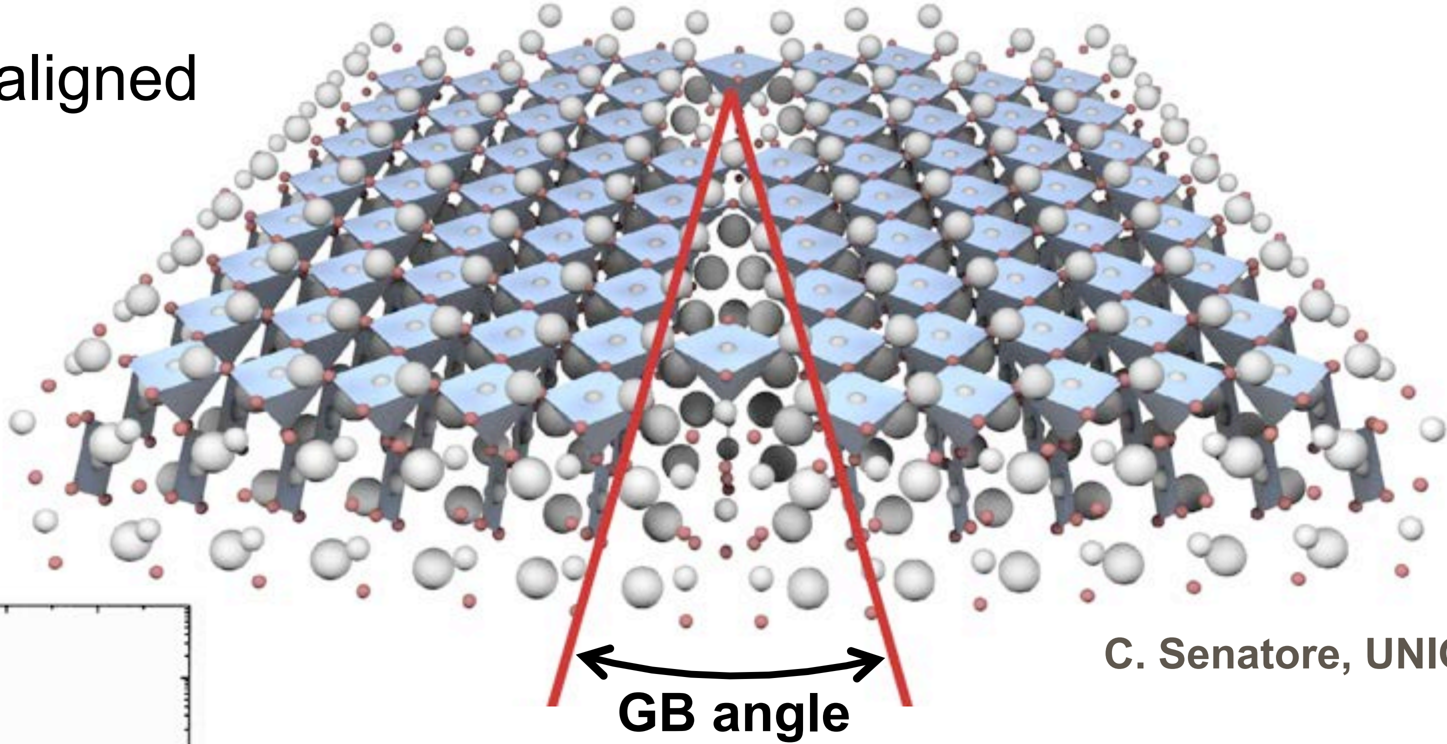



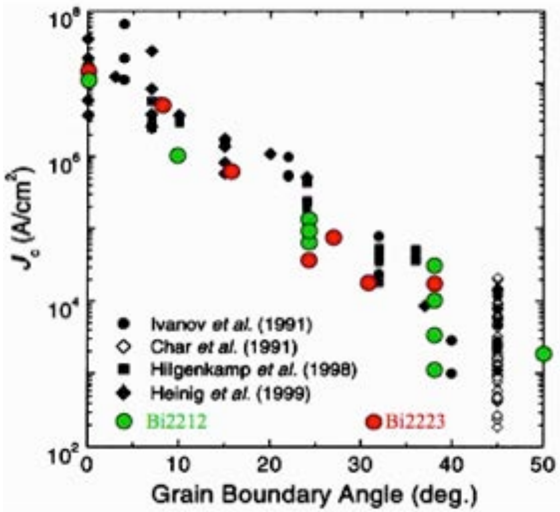

**$J_c$ (77 K, 0T)**

- Ceramic brittle materials
- Very sensitive to defects ($\xi$)

# HTS specificities

ab planes aligned




ab planes alignement not sufficient!

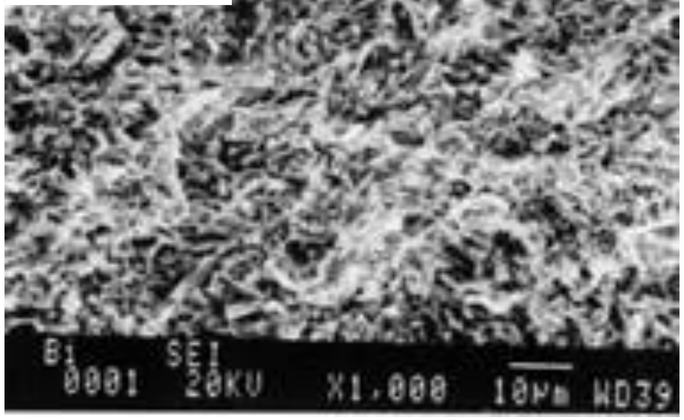
Deviations higher than 6 degrees already reduce $J_c$ significantly

Challenge: make a „crystal“ on km lengths!

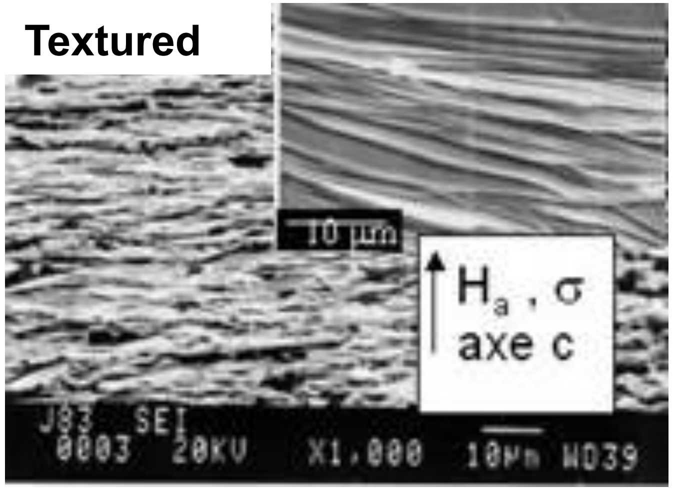
# Texturation

- Mechanical texturation (drawing): 1G BSCCO

Sintered




Textured



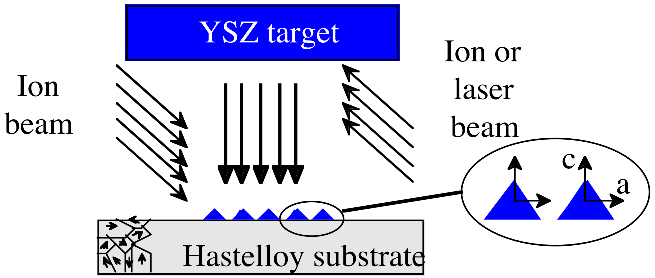

- Texturation by epitaxial deposition: 2G YBCO
  - Requires to deposit on a textured support

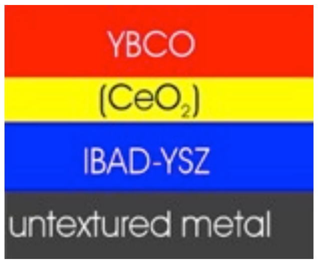
*Coated conductors*

# Coated conductors - two main techniques

*Basis: deposition on a textured support*
*Textured layer on a substrate or textured substrate*

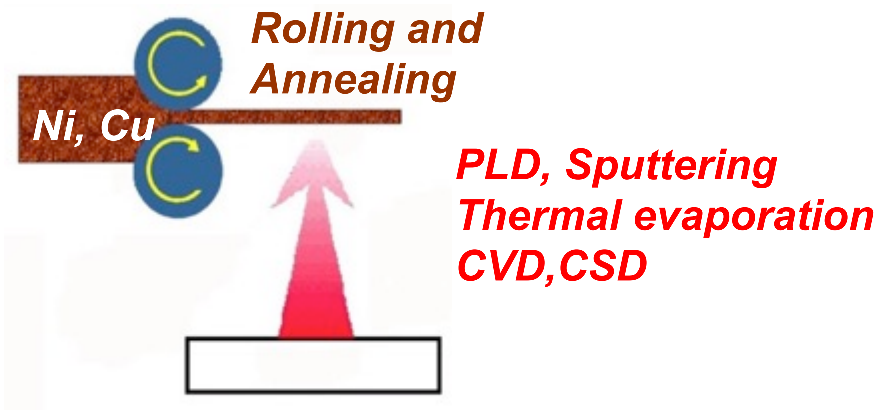
### Ion Beam Assisted Deposition (IBAD)




Textured buffer layer

### Rolling Assisted Biaxially Textured Substrates (RABiTS)

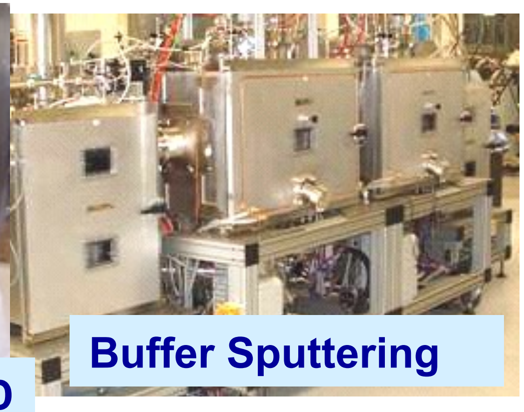


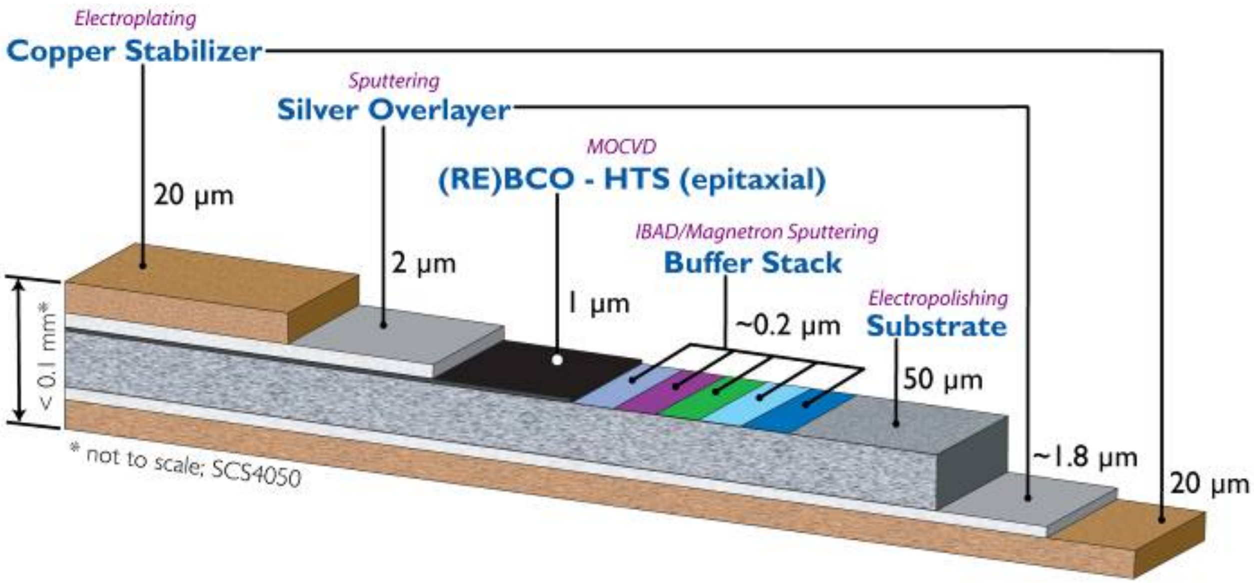


Textured substrate

# Coated conductors – IBAD process



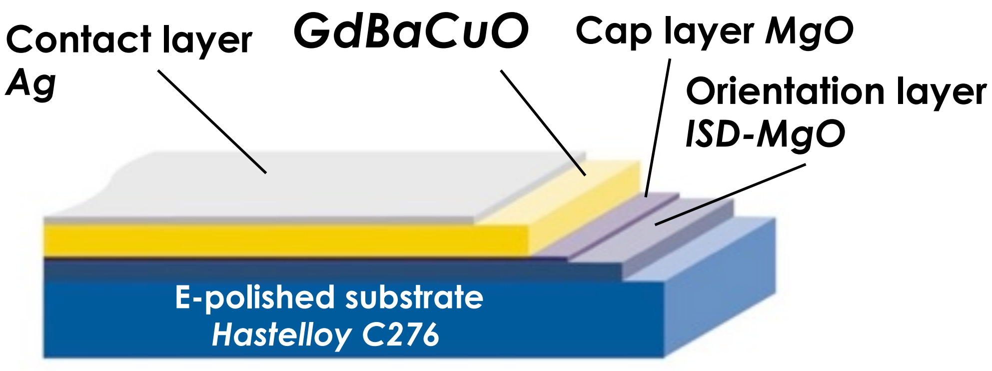

Deposition methods

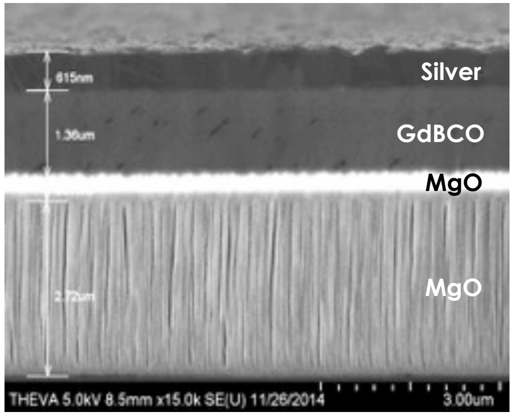
- Physical
- Chemical

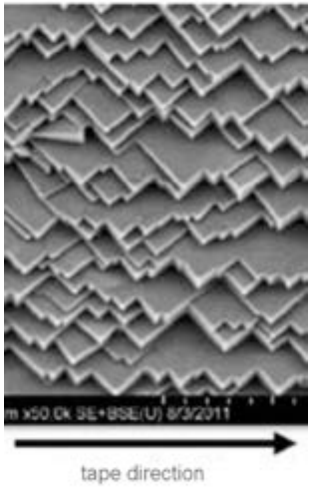
SuperPower Inc.
A Furukawa Company

# Coated conductors

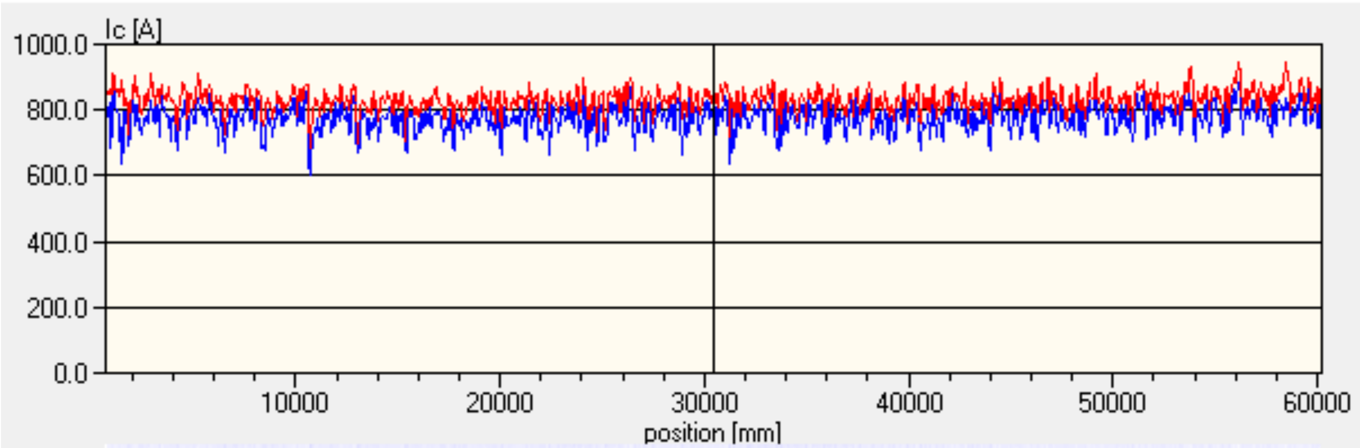
THEVA

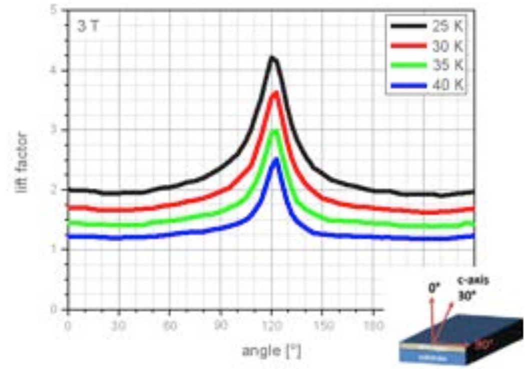



MgO surface




$I_c \approx$ **700 A/cm-w (77 K, sf)**

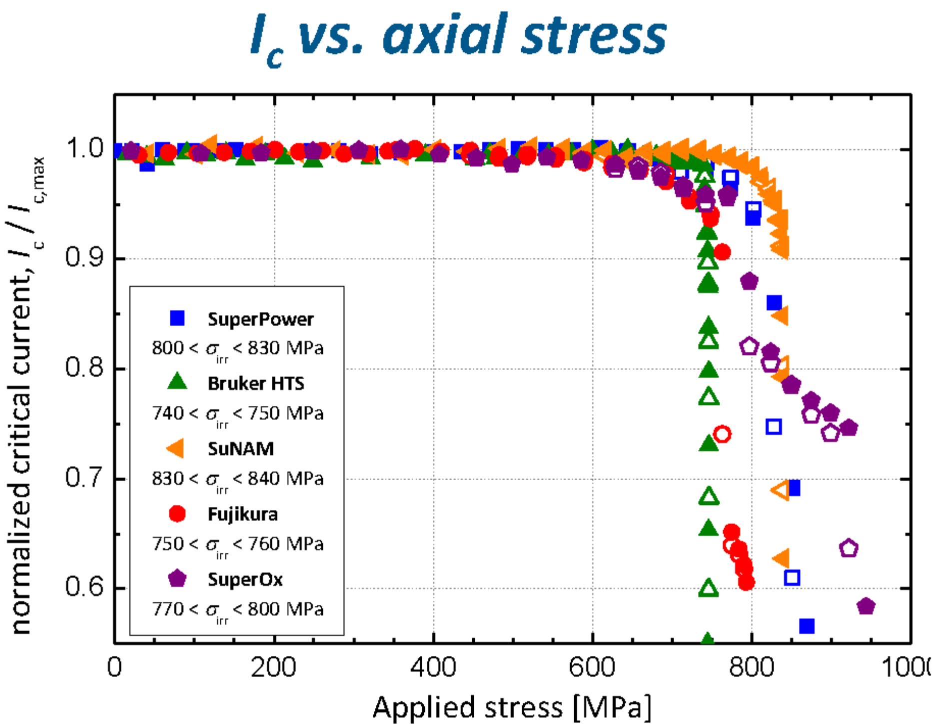
Inclined Substrate Deposition (ISD) process

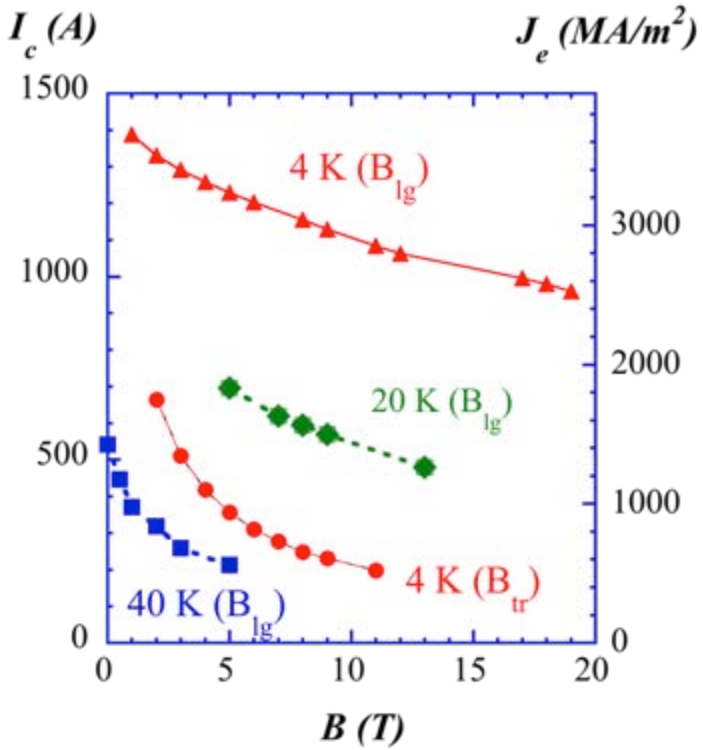


UNIVERSITÉ Grenoble Alpes

# Coated conductors

## Typical critical characteristics




*Performances fit main applications*
*Really technical superconducting conductor*

# HTS materials summary

Cost, cost and cost!

- PIT: niche applications, a few manufacturers
- $MgB_2$: niche applications (pending CC?)
- CC: large hopes, a lot of manufacturers

***A lot of room for improvements***

**HTS applications**

- First application: HTS current leads
- Second application: FCL, HTS high field insert

# Superconducting materials

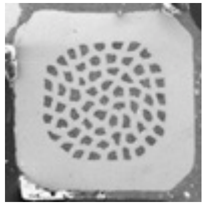
**Low $T_c$**

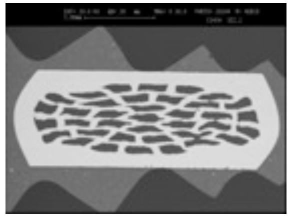
- **NbTi (3000 t/an)**

- **$Nb_3Sn$ (15 t/an)**

---

- **$MgB_2$**





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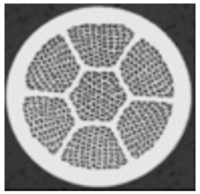
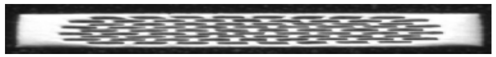
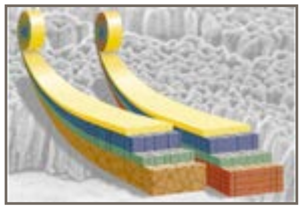
**High $T_c$**

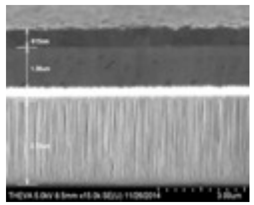
- **BiSrCaCuO (1G)**





- **YBaCuO (2G)**



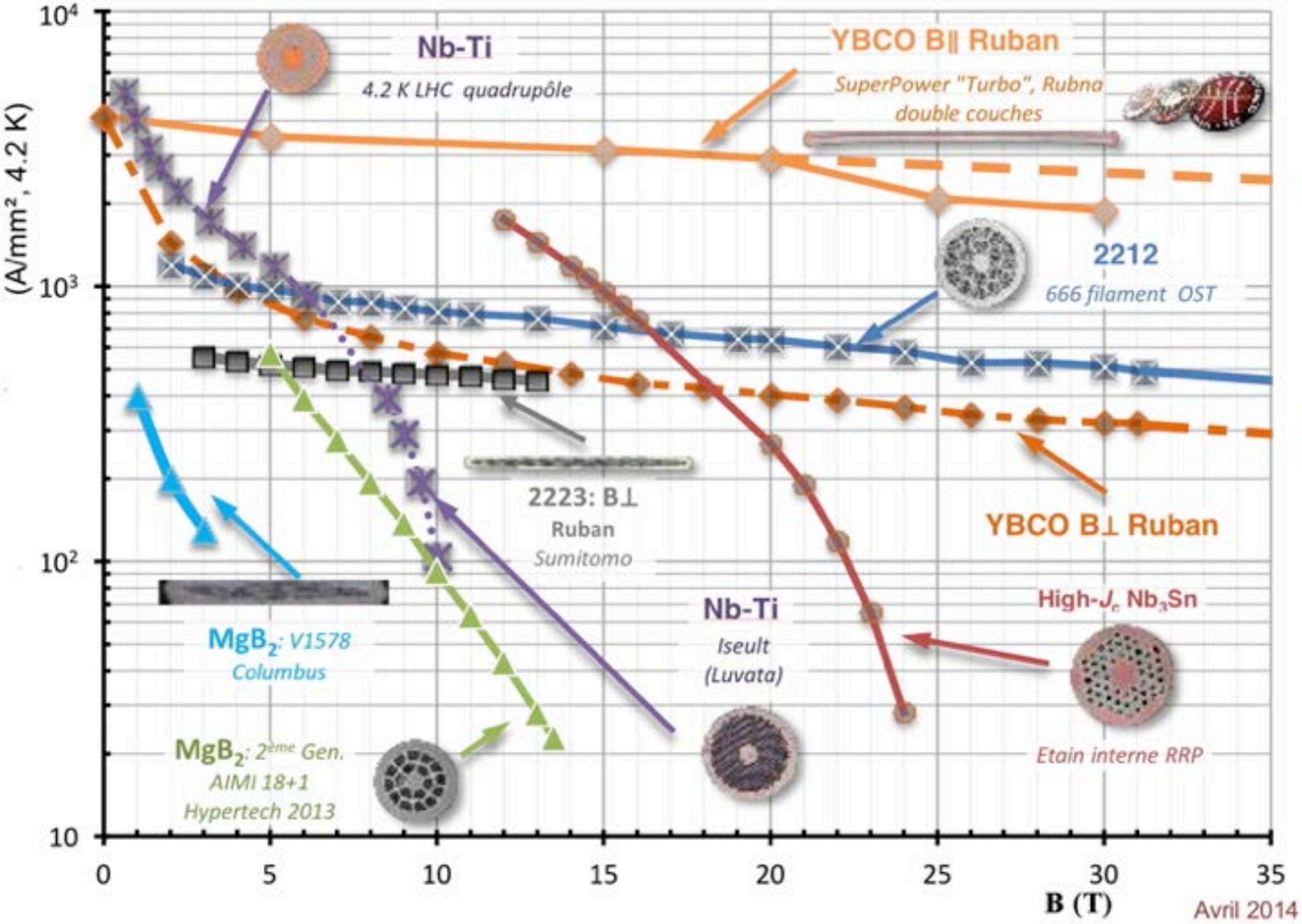


# Superconducting materials

| Material | State | Length | Cost* | Applications |
|---|---|---|---|---|
| **NbTi** | Mature | / | Low (1) (< Cu) | < 9 T (4 K) < 12 T (1,8 K) |
| **$Nb_3Sn$** | Still progresses | / | 5 - 10 | > 12 T |
| **$MgB_2$** | R & D Margins | km | 5 – 10 | < 3 T @ 20 K |
| **BSCCO (1G)** | R & **D** Progress | km | 50 – 100 (Ag) | (inserts) |
| **YBCO (2G)** | **R & D** Cost | Hundred m | 100 – 250 (process) | Inserts |

*** : €/kA/m (77 K, 0 T, HTS), (4 K, $B_o$, LTS)**

# Superconducting materials



# Summary HTS – Take home message

- HTS devices still very small market (30 M$ / 5050 M$)
- HTS: 2 wire generations
  - 1G: BSCCO tape and round wire (niche market)
  - 2G: YBCO coated conductor (great hopes)
- 1G: niche market due to high intrinsic cost (Ag)
- 2G: low material cost, high process cost
  - High performances ($J_{ce}$, mechanics)
  - Anisotropy, delamination, cable
- $MgB_2$: low cost, competition to LTS for medium fields at about 20 K, DC applications, advances underway

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# Superconductivity II
## Superconducting Application Introduction


Pascal Tixador

Grenoble-INP, Université de Grenoble Alpes, CNRS
G2ELab, Institut Néel

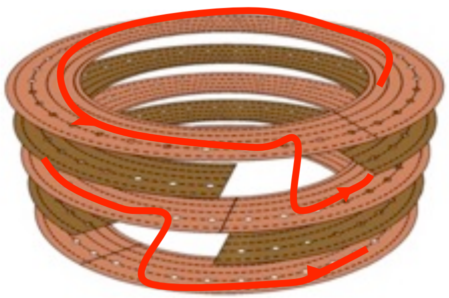

UNIVERSITÉ Grenoble Alpes   Grenoble INP   CNRS   NÉEL institut   G2E Lab Grenoble Génie Electrique Grenoble Electrical Engineering

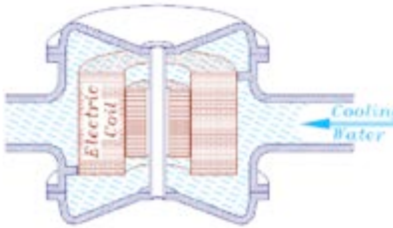
# Superconductor interests

- High current densities under magnetic fields
  - Volume reduction
- No loss in DC environment
  - Energy savings
  - Possible energy storage
  - Magnetic bearings
- SC to dissipative state transition
  - Fault current limitation

# Comparison for a magnet

| | Bitter | Supercon. | Ratio |
|---|---|---|---|
| Magnetic flux density (T) | 10 | 11 | 1.1 |
| Core (mm) | 500 | 500 | 1 |
| Supply (kW) | 14 000 | 5 | 2 800 |
| Cooling (kW) | 200 (pumps) | 100 (liquefier) | 2 |






# Losses in SC (under critical surf.)

Function of the electromagnetic environment

- Time constant electromagnetic environment
  - No or very low losses
- Time variable electromagnetic environment
  - Losses: AC losses

$$\overrightarrow{curl}\,\vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \vec{E} + \vec{J} : losses\left(\delta P = E\,J\right)$$

**=> Interest more in DC than AC**

# Superconductor uses

**Applications with large stored magnetic energy**

$$W_{mag} = \frac{1}{2} \iiint_{Espace} B \cdot H \; dxdydz = \frac{1}{2\,\mu_o} \iiint_{Espace} B^2 \; dxdydz$$

- Applications difficult without superconductors
- New types of applications
- Improvment of conventional applications

# Applications difficult without SC

Superconductivity inescapable

Devices difficult to design without superconductivity

- High magnetic flux densities
- Large magnetized volumes

| MRI | RMN Spectrometer | Magnets | High energy physics | Thermonuclear fusion |
|---|---|---|---|---|

Philips©

30 000

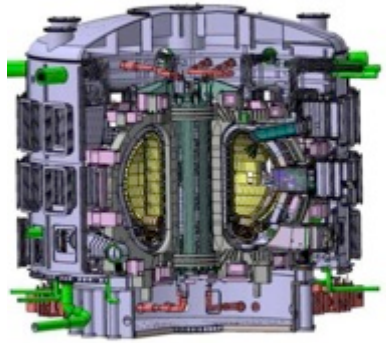
Bruker©

CNRS©

Cern©

27 km (100)

40 MW (900)



ITER©

78 %

20 %

# New types of applications

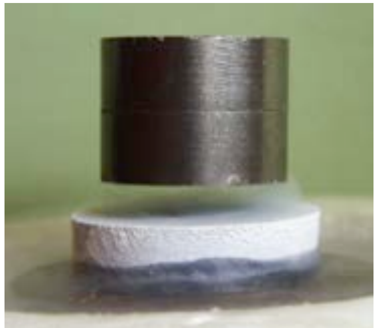
New functions

Innovative devices

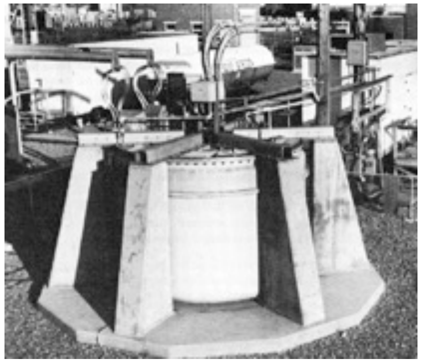
Properties inherent to superconductors

- quench from SC to normal state
- No DC losses

| Magnetic bearings | Fault Current Limiter (FCL) | SMES (magnetic storage) |
|---|---|---|



NEXANS©



IEEE©

# Improved devices

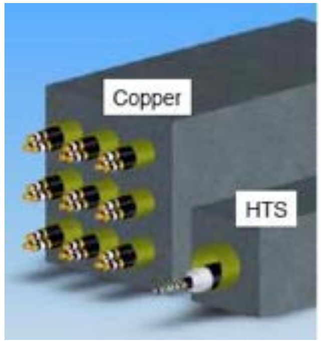
Improvement of resistive devices

- Enhanced compactness & efficiency, weigth reduction
  - no DC loss

+

Possibility to integrate innovative fonctions (limitation)

| Electrical rotating machine | Cables | Transformers |
|---|---|---|

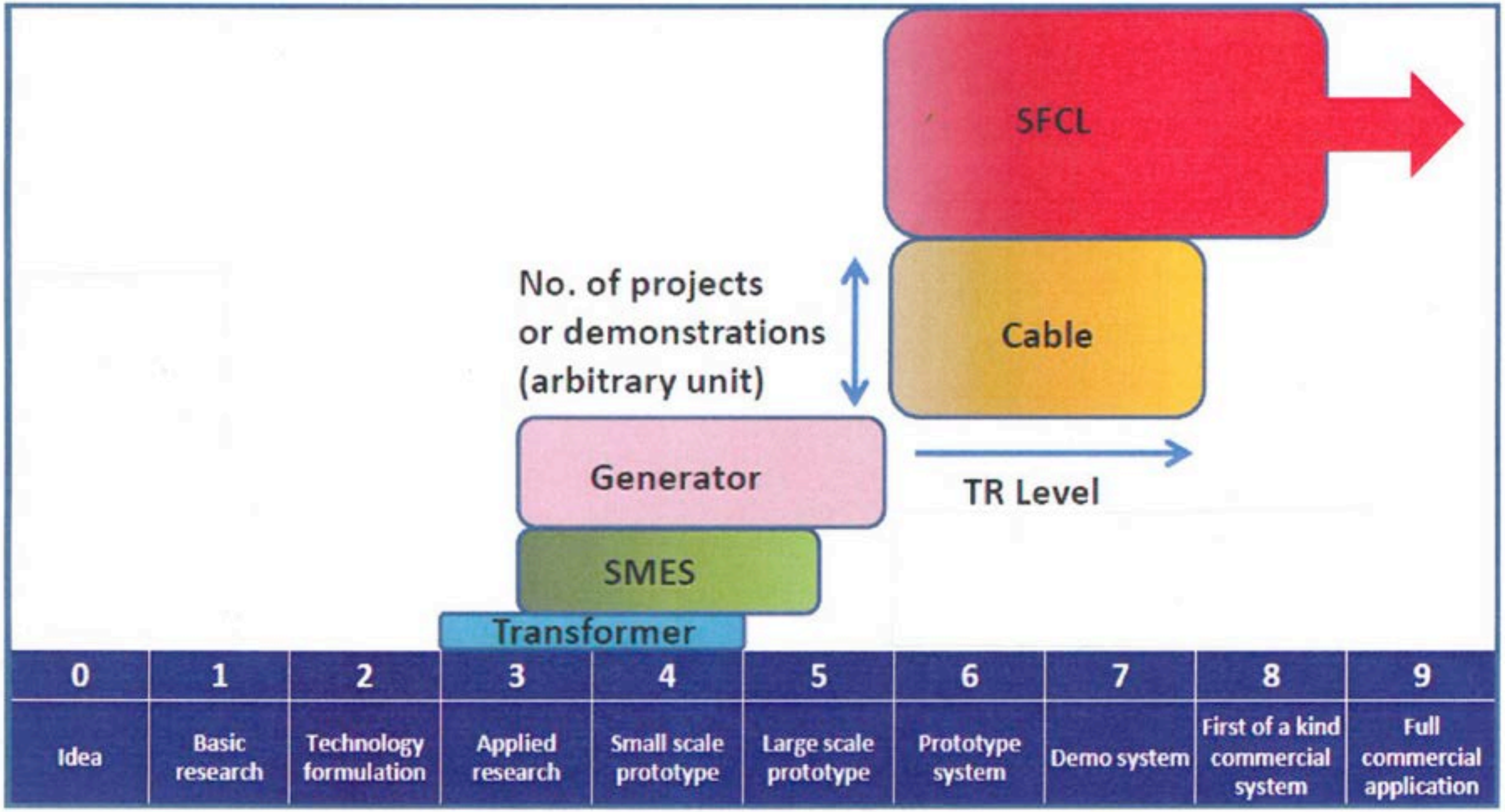




Siemens©

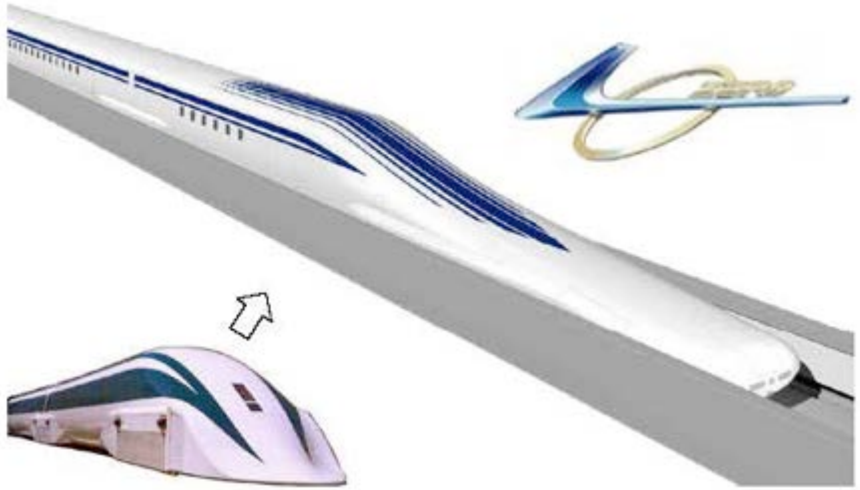
# Supercond. and large scale applications

- Higher power Quality and security
  - FCL, Low impedance cable, high short-circuit power, high meshing
- Higher power density
  - Transfo. & machine (30 / 50 %), cable (> 50 %)
- Enhanced efficiencies
  - Transfo. (50 % (> 80 % mobile)), machines (30/50 %) magnets (> 90 %)

# SC power application - Summary



# Maglev

505 km/h

Project approved by Japan:

- Tokyo-Nagoya 2027 – 286 km – 40 min (5.5 trillion yen)
- Tokyo-Osaka 2045 – 438 km – 67 min (9 trillion yen total 65 b€)

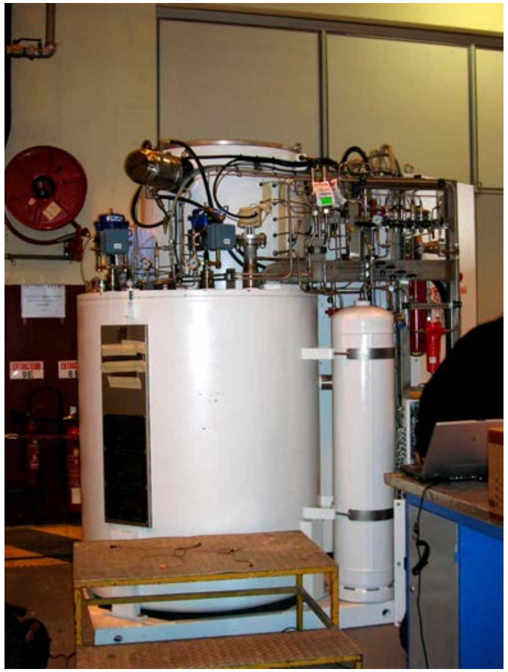
Tests (Yamanashi test line - 42.8 km): > $10^6$ km

# Superconducting power devices

- Only very few industrial devices
  - NbTi magnets for magnetic separation
  - A very few FCL
  - Many very successful experiments
- Three main reasons
  - Resistive systems in general high performances
  - Cryogenic cost and **fear**
  - SC device: revolution, not simple evolution

**Superconducting power devices economically possible only with HTS at moderate cost**

# Do not forget



250 W @ 20 K

AIR LIQUIDE
TECHNIQUES AVANCEES

250g – 0.2 W @ 77 K


200 kV bushing

Fields perfectly mastered

*if correctly designed!*

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# Superconducting Fault Current Limiter (FCL)

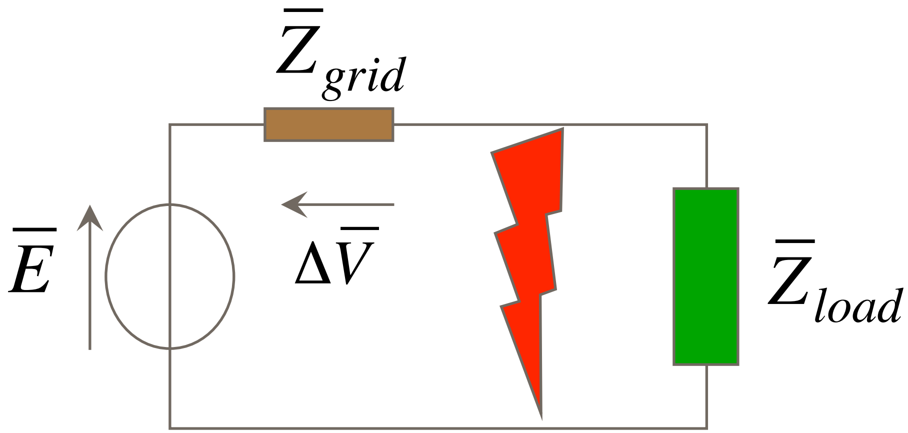
Pascal Tixador

Grenoble-INP, Université de Grenoble Alpes, CNRS

G2ELab, Institut Néel

# Fault Current Limiter (FCL) - Need

## Ultra simple representation of a grid:




Grid design: compromise

- Power quality low $Z_{grid}$
  - ✓ $\Delta V$
  - ✓ Perturbation

Fault current $E/Z_{grid}$

- ✓ Device over-design
- ✓ Breaker cutting capacity

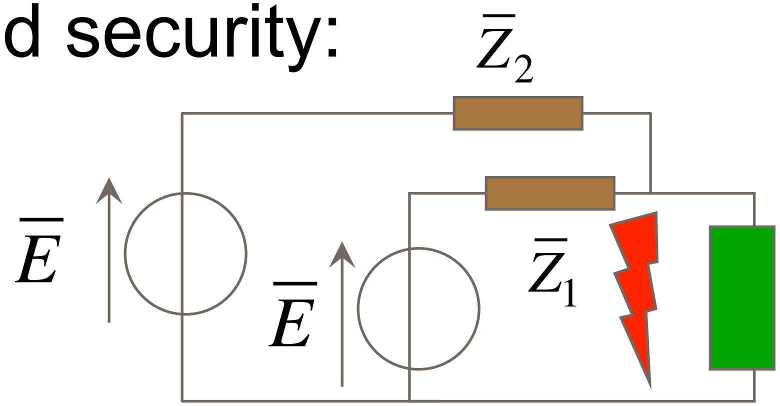
# Need for a fault current limiter

Electric grids:

- extremely important in our countries
- critical infrastructures

Security and power quality:
two main demands today


Improved security:




Effective but if fault:

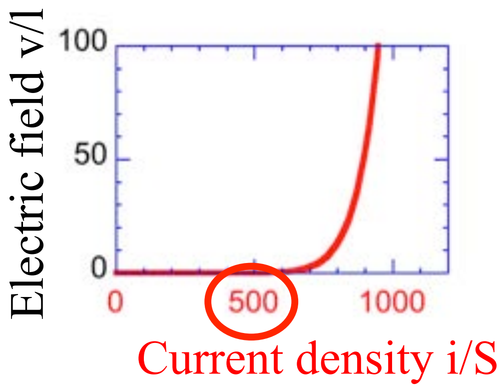
$I_{cc} = I_{cc1} + I_{cc2}$

FCL required

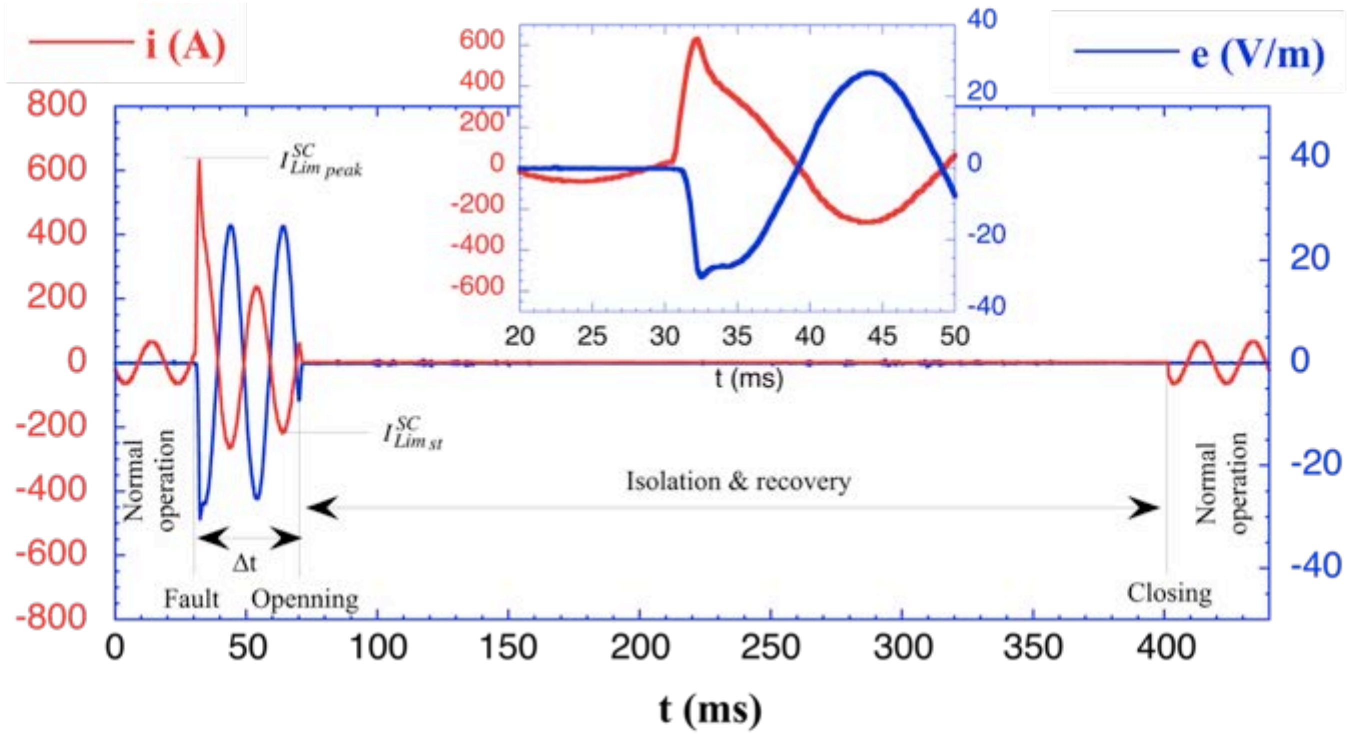
No today satisfying FCL

*Superconductivity brings attractive response*

# SC FCL

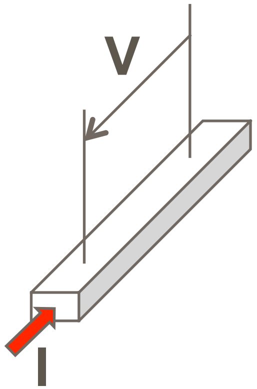


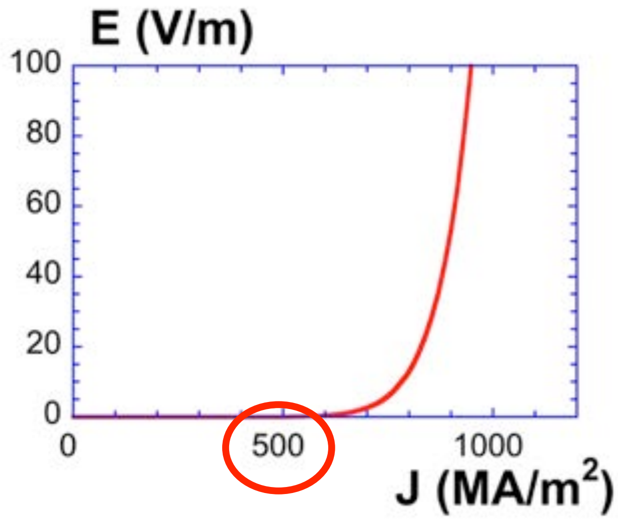


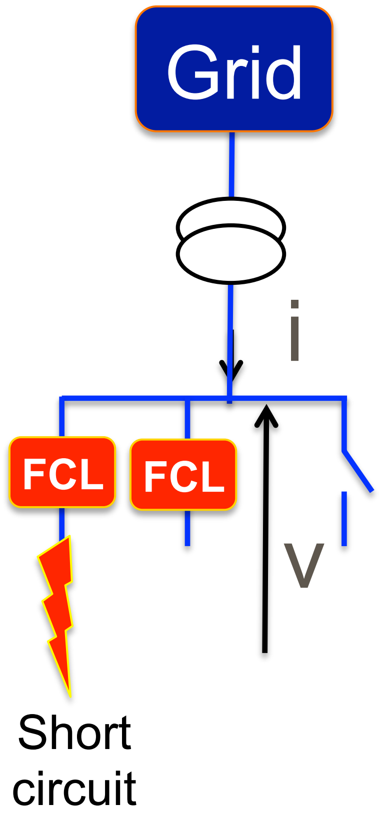


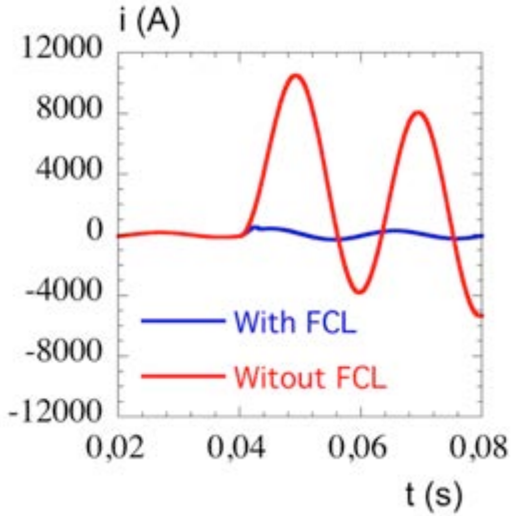

FCL makes possible a low grid (zero!) impedance BUT a low fault current. Graal for grid designers!

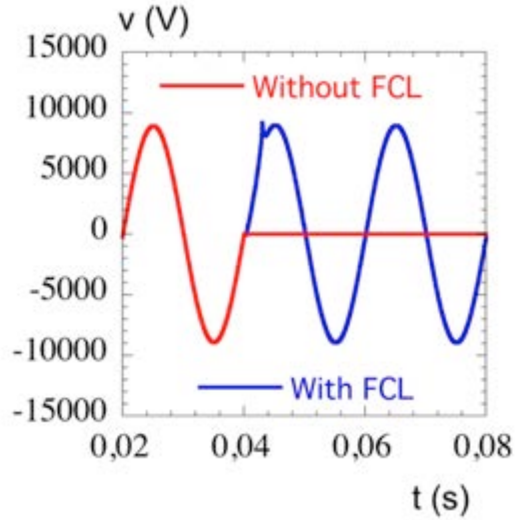
# Fault Current Limiter - FCL




SFCL
- Nearly invisible in NO
- Self triggering
- Fail safe
- Automatic recovery

Uses
- Current limitation
- Fault « isolation »

# FCL – State of the art


(Courtesy L. Martini)

Italy 9 kV / 0.2 kA

(Courtesy Nexans)

Germany 12 kV / 0.8 kA


(Courtesy H.-R. Kim )

Korea 23 kV / 0.63 kA

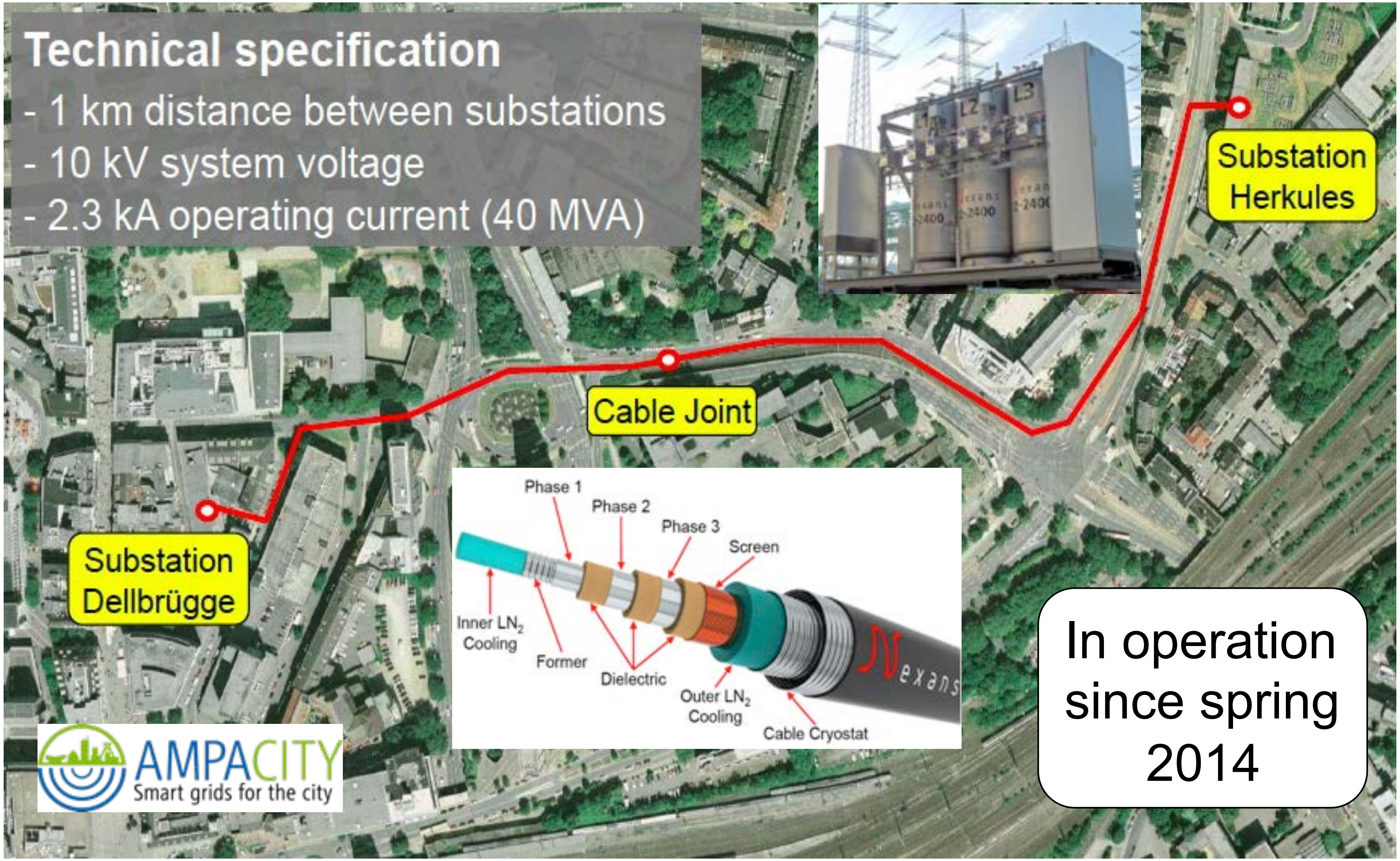
China 220 kV / 0.8 kA

APPLIED MATERIALS®

USA (Santa Clara) 15 kV / 1 kA

Many successful experiences.
Some test experiences

# AMPACITY project: cable + FCL




Courtesy M. Stemmle

# FCL from SIEMENS in Augsburg

## „ASSiST“ – SFCL for public 10 kV grid of Stadtwerke Augsburg, Bavaria, Germany

swa Netze SIEMENS

SFCL based Solution (SFCL+switchgear+control+DAQ) successfully installed and inaugurated in 03/2016.

gefördert von Bayerisches Staatsministerium für Wirtschaft und Medien, Energie und Technologie

**Details**

- Collaboration of Siemens EM, CT & Stadtwerke Augsburg
- Integration of MTU's extended testing facility of combined heat and power unit requires reduction of short-circuit current
- Combination of superconducting 15 MVA SFCL with ultrafast breaker and parallel series reactor
- Closed cooling system (cold heads included) – no blow-off during limitation
- Reduction of losses compared to conventional solution
- Increased system stability, no voltage drop
- Large area breaker up-grade dispensable
- Timeline:
  - Apr. 15, 2014: project start
  - Mar. 15, 2016: official inauguration
  - till Jan. 15, 2017: data acquisition & monitoring (to Siemens Erlangen)
  - Continued operation planned after project end

Testing at CT, Erlangen

Installation area


Installed vessels

Inauguration ceremony

Courtesy Tabea Arndt - Siemens

# FCL: market starts! Birmingham (UK)

Commercial SCFCL in Western Power Distribution (UK)

Bus tie coupling

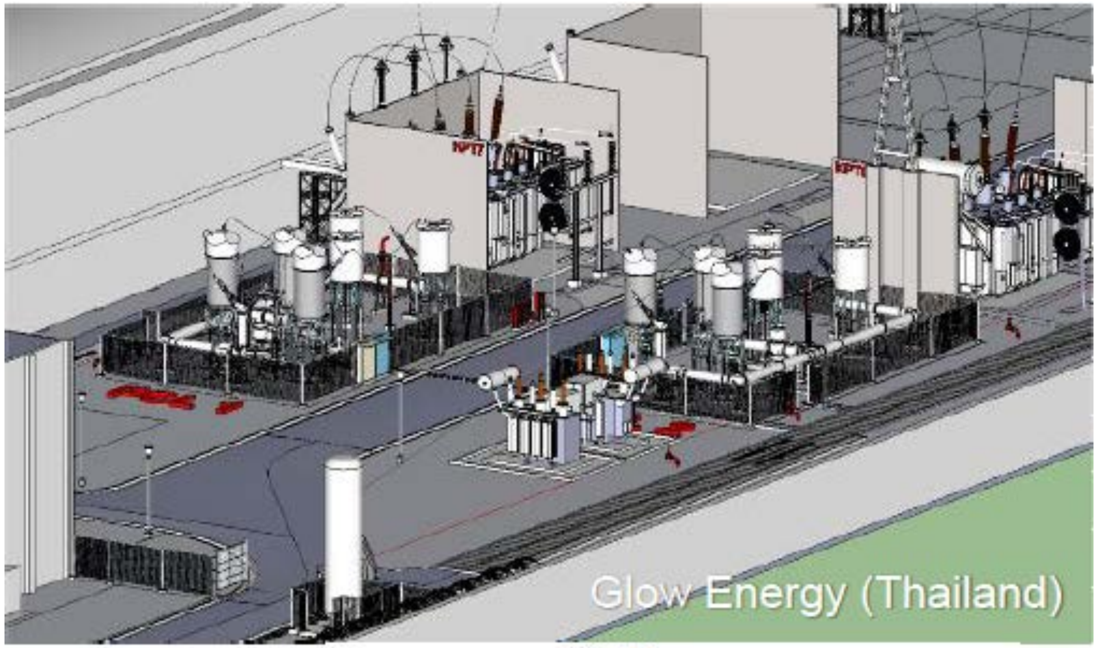
| Location | Chester | Bournville |
|---|---|---|
| $U_a$ | 12 kV | 12 kV |
| $I_a$ | 1.6 kA | 1.05 kA |
| $I_{pros}$ | 20 kA peak | 22 kA peak |
| $I_{lim}$ | 9.9 kA peak | 7.7 kA peak |
| $\Delta t$ | 100 ms | 100 ms |
| Recovery | 30 s | 30 s |
| Location | Outdoor | Indoor |
| Cooling power (RT) | 18-50 kW | 18-30 kW |

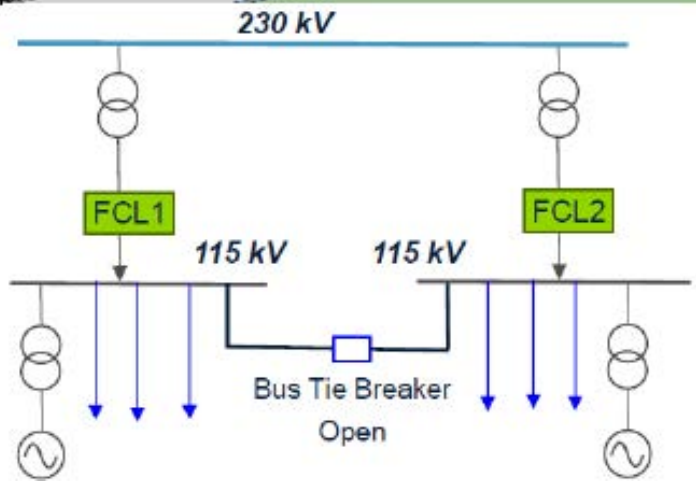

SCFCL put in operation late 2015

WESTERN POWER DISTRIBUTION

SCFCL 12 kV / 1.6 kA Starting end 2015

# FCL: market starts! Rayong (Thailand)




APPLIED MATERIALS®

| | |
|---|---|
| Voltage | 115 kV |
| Rated current | 550 A |
| Iprospective | 5 kA |
| Ilim | 2.5 kA |
| Shunt (react) | 15 Ω |
| BIL | 550 kV |
| AC withstand | 230 kV |

# SCFCL- Conclusions

- SCFCL: an innovative device for a real demand (better grids)
  - High short-circuit power but low fault currents (Graal!)
- First commercial SCFCLs in MV (Birmingham) and HV (Thailand)
- SCFCLs already out of the laboratory development phase
- Now midway between the pilot projects & industrial deployment
- Many people still think SFCL remains a laboratory device
  - Communication about field test experiences
- Grid designers should know that they have a new device
- Have to suppress the psychological fear against cryogenics
- Field tests on long periods to remove last utility's fears
- A lot of space for improvements
- Conductor cost should no more be an issue
- Cryogenic could be critical for cost

# Conclusion, take home message

- LTS market: mature niche market still growing
  - MRI, NMR (physics)
  - 4 K cryogenics remains an economical hurdle

- HTS market emerges
  - Cryogenics no more an economical hurdle
  - Conductor remains today an economical hurdle
  - Lot of advances in YBCO, many actors
  - Superconductivity: enabling technology
    - SC FCL, VLI cable, rot. mach. for wind farm, planes…
  - HTS market takes off with a few FCL orders
  - Fears (cryogenics) should be overcome
    - Field tests on long periods to remove last utility's fears
  - Communications
    - SC FCL exist: a new device for grids

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# Gràcies!